An Introduction to Fractal Geometry, Illustrated by the Mandelbrot Set

2026 edition with further chapters

by

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Preface

This *Facharbeit*, "An Introduction to Fractal Geometry, Illustrated by the Mandelbrot Set", was prompted by an article in the German-language edition of "Scientific American". That publication presented the background to an object of recreational mathematics in an accessible form and showed fascinating computer graphics.

The aim of this Facharbeit is to clarify the mathematical framework behind such complex structures. The structures themselves conceal deep mysteries, and their investigation is constantly advancing.

It is obvious that a Facharbeit can treat the subject only to a limited extent, since this is a relatively new branch of discrete mathematics and the mathematical relationships are more complicated than they are often made out to be.

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Note on the new edition 2026

This edition was produced in 2026 on the basis of the 1990 Facharbeit. It follows the original's structure and line of argument, but has been reworded and considerably shortened: many calculations, examples, figures and individual sections of the original have been condensed or omitted, and mathematically incorrect statements have been corrected. Only some of these changes are marked individually in the highlighted note boxes and in the change log.

The 1990 original is preserved, together with a verbatim transcription and a facsimile of every page, in a private archive. The figures were recalculated or redrawn in 2026. The Python code used for this is documented in the appendix, separately from the original 1990 Pascal code.

Chapters 5 to 9 ("Further topics") were added in 2026. They build on chapters 1 to 4 and lead from the Mandelbrot set to Julia and Cantor sets, Hausdorff dimension, other complex functions and open questions in current research.

1 Introduction to complex numbers

1.1 General remarks

In algebra, when solving equations, for example when finding the zeros of a function, one frequently encounters the problem that an equation has no real solution. Within the real numbers, the square root of a negative number cannot be taken. This restriction can be overcome by extending the number system and introducing a new number i with the property $i^2 = -1$.

1.2 Definition of the complex numbers

A complex number consists of a real component and an imaginary component. The set of complex numbers is defined by

$$\mathbb{C} = \{\, a + bi \mid a,b \in \mathbb{R} \,\}, \qquad i^2 = -1.$$

For $z = a + bi$, $a = \mathrm{Re}(z)$ is called the real part and $b = \mathrm{Im}(z)$ the imaginary part. The notation $i = \sqrt{-1}$ is to be understood only as an intuitive shorthand; the actual definition is $i^2 = -1$.

1.3 The four basic arithmetic operations

Let $z_1 = a + bi$ and $z_2 = c + di$ with $a,b,c,d \in \mathbb{R}$.

1.3.1 Addition and subtraction

$$z_1 + z_2 = (a + c) + i(b + d), \qquad z_1 - z_2 = (a - c) + i(b - d).$$

Complex numbers are added or subtracted by adding or subtracting their real parts and their imaginary parts separately.

1.3.2 Multiplication

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + i(ad + bc).$$

Here the brackets are multiplied out as with real expressions, and $i^2 = -1$ is then substituted. For the complex conjugate $\overline{z} = a - bi$ we have, in particular,

$$z\overline{z} = (a + bi)(a - bi) = a^2 + b^2.$$

1.3.3 Division

For $z_2 \neq 0$, that is $c^2 + d^2 \neq 0$, the numerator and denominator of the fraction are multiplied by the complex conjugate of the denominator:

$$\frac{z_1}{z_2} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{ac + bd}{c^2 + d^2} + i\,\frac{bc - ad}{c^2 + d^2}.$$

Both components of the result are real numbers.

1.4 Basic laws of algebra in $\mathbb{C}$

The arithmetic laws familiar from the real numbers also hold for complex numbers. For $z_1,z_2,z_3 \in \mathbb{C}$ we have the commutative law

$$z_1 + z_2 = z_2 + z_1, \qquad z_1 z_2 = z_2 z_1,$$

the associative law

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), \qquad (z_1 z_2) z_3 = z_1 (z_2 z_3),$$

and the distributive law

$$z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3.$$

The identity elements are $0 = 0 + 0i$ for addition and $1 = 1 + 0i$ for multiplication. The additive inverse of $z = a + bi$ is $-z = -a - bi$. For $z \neq 0$ the multiplicative inverse also exists:

$$z^{-1} = \frac{\overline{z}}{z\overline{z}} = \frac{a - bi}{a^2 + b^2}.$$

The complex numbers with addition and multiplication therefore form a field.

> **Correction note** In the original, individual transformations were worked through in detail, but the identity elements and the multiplicative inverse were not stated in full. This gap, marked by the teacher, has been closed here.

1.5 The modulus (absolute value) in $\mathbb{C}$

The modulus of a complex number $z = a + bi$ is defined by

$$|z| = |a + bi| = \sqrt{a^2 + b^2}.$$

The following hold:

$$|z| \geq 0, \qquad |z| = |-z|, \qquad |z_1 z_2| = |z_1|\,|z_2|$$

and, for $z_2 \neq 0$,

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}.$$

In addition, the triangle inequality holds:

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

To prove it, one can square both non-negative sides. With $z_1 = a + bi$ and $z_2 = c + di$ it then suffices to show

$$ac + bd \leq \sqrt{a^2 + b^2}\sqrt{c^2 + d^2},$$

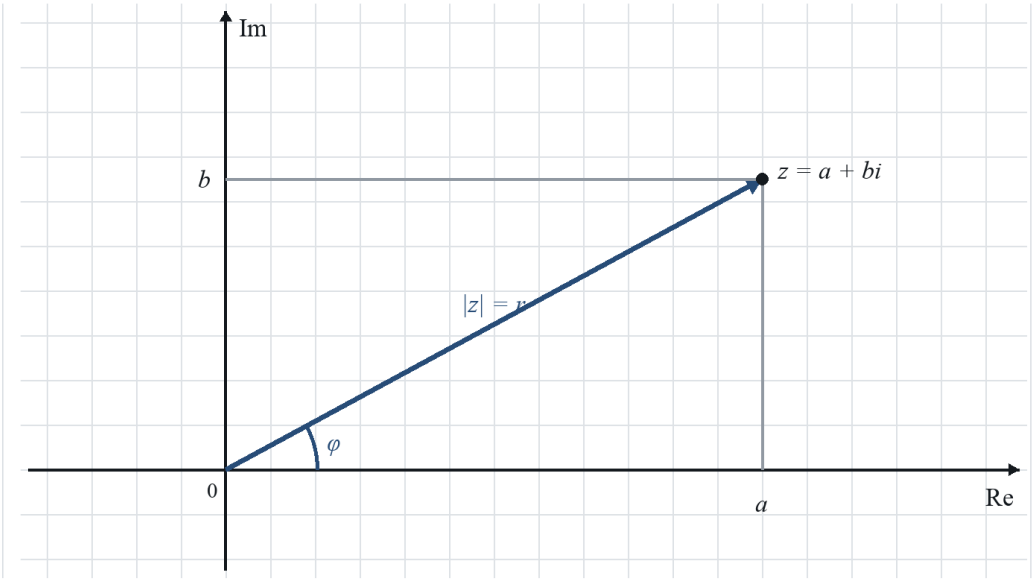
which is the Cauchy–Schwarz inequality in $\mathbb{R}^2$. Equivalently, the transformation used in the original ends with the true statement

$$0 \leq (ad - bc)^2.$$

In the complex plane (Gaussian plane), $z = a + bi$ corresponds to the point (a,b). The modulus $|z|$ is its distance from the origin, and $|z_1 - z_2|$ is the distance between the corresponding points. The conditions $|z| = 1$, $|z| < 1$ and $|z| > 1$ describe the unit circle, its interior and its exterior.

1.6 Geometric interpretation

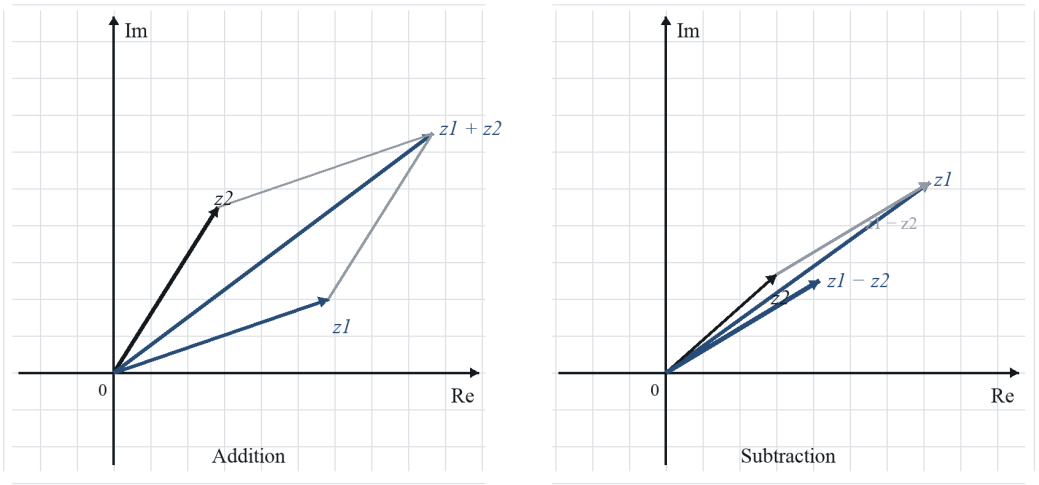
The real part is plotted on the horizontal axis of the complex plane and the imaginary part on the vertical axis. Addition and subtraction can thus be represented as vector addition and vector subtraction respectively.



The point $z = a + bi$ is determined by its two coordinates a and b. Equivalently, it can be described by its distance $r = |z|$ from the origin and the angle φ to the positive real axis.

1.6.1 Addition and subtraction

Every complex number can be regarded as a position vector. Completing the vectors to a parallelogram gives the sum of two complex numbers. The difference $z_1 - z_2$ is obtained by adding z_1 and $-z_2$.



When adding, $z_1 + z_2$ arises as the diagonal of the parallelogram. The difference $z_1 - z_2$ can be interpreted as the vector joining z_2 to z_1.

1.6.2 Multiplication and division

For $z \neq 0$ we introduce polar coordinates. With $r = |z| > 0$ and an argument φ we have

$$z = r(\cos\varphi + i\sin\varphi).$$

For

$$z_1 = r_1(\cos\varphi_1 + i\sin\varphi_1), \qquad z_2 = r_2(\cos\varphi_2 + i\sin\varphi_2)$$

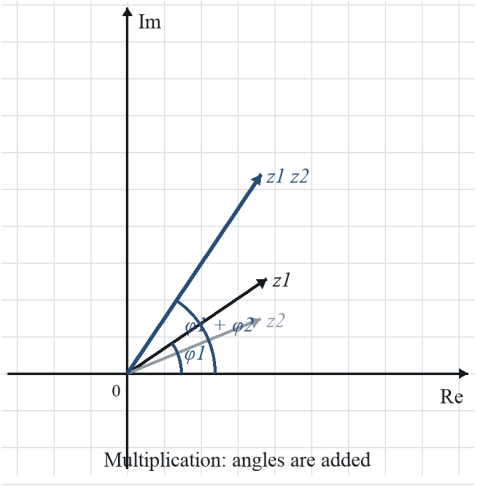
the angle-addition formulas give

$$z_1 z_2 = r_1 r_2\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big).$$

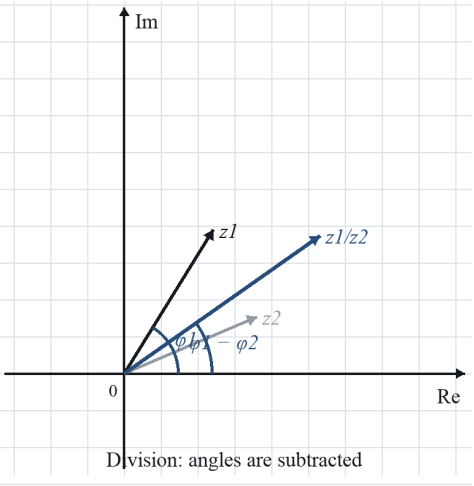
In multiplication, therefore, the moduli are multiplied and the arguments added. Correspondingly, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\big(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)\big).$$

In division, the moduli are divided and the arguments subtracted.



Multiplication: angles are added



Division: angles are subtracted

The drawings illustrate only the angle relationships. The length of the product vector or of the quotient vector also depends on the moduli $r_1 r_2$ or r_1/r_2 respectively.

1.7 Ordering of complex numbers

The complex numbers cannot be ordered in such a way that, together with the usual arithmetic operations, they form an ordered field. If $i > 0$, then compatibility with multiplication would give $i^2 = -1 > 0$. If, on the other hand, $i < 0$, then $-i > 0$, and likewise $(-i)^2 = -1 > 0$ would follow. Both contradict $1 > 0$.

As a mere set, $\mathbb{C}$ can of course be ordered, for instance lexicographically by real and imaginary part. Sorting by modulus and then by a fixed choice of argument is also possible. Such orderings, however, are not compatible with addition and multiplication in the way that the usual ordering of the real numbers is.

Correction note The original proof considered only the case $i > 0$, and the subsequent ordering by modulus and angle was not fully specified. Both of the teacher's remarks have been taken into account here.

1.8 Examples

Example 1 The equation $z^2 + 2 = 0$ has the two solutions

$$z_1 = i\sqrt{2}, \qquad z_2 = -i\sqrt{2}.$$

Example 2 For $z^4 + 3 = 0$, that is $z^4 = -3$, we write

$$-3 = 3\bigl(\cos(180^\circ) + i\sin(180^\circ)\bigr).$$

De Moivre's theorem gives the four solutions

$$z_k = \sqrt[4]{3}\left(\cos\frac{180^\circ + 360^\circ k}{4} + i\sin\frac{180^\circ + 360^\circ k}{4}\right), \qquad k = 0,1,2,3.$$

Their arguments are 45°, 135°, 225° and 315°.

Example 3 For $z^5 + 4 = 0$, that is $z^5 = -4$, we correspondingly have

$$z_k = \sqrt[5]{4}\left(\cos\frac{180^\circ + 360^\circ k}{5} + i\sin\frac{180^\circ + 360^\circ k}{5}\right), \qquad k = 0,1,2,3,4.$$

The five arguments are 36°, 108°, 180°, 252° and 324°. The value 342° given in the original was a slip of the pen.

2 Sequences and limits in $\mathbb{C}$

2.1 Sequences and convergence in $\mathbb{C}$

A complex sequence is a mapping

$$\mathbb{N} \longrightarrow \mathbb{C}, \qquad n \longmapsto z_n.$$

Writing $z_n = a_n + ib_n$, the real and imaginary parts each form a sequence of real numbers. A complex sequence is called convergent if the sequence of real parts and the sequence of imaginary parts both converge.

2.1.1 Definition of the limit

The sequence (z_n) converges to $z \in \mathbb{C}$ if for every $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that

$$|z_n - z| < \varepsilon \qquad \text{for all } n \geq N.$$

One writes

$$\lim_{n\to\infty} z_n = z.$$

For $z_n = a_n + ib_n$ and $z = a + ib$,

$$z_n \longrightarrow z \quad \Longleftrightarrow \quad a_n \longrightarrow a \ \text{ and } \ b_n \longrightarrow b.$$

From $|z_n - z| < \varepsilon$ it follows immediately that $|a_n - a| < \varepsilon$ and $|b_n - b| < \varepsilon$. Conversely, one chooses the bound $\varepsilon/2$ for both real sequences; then

$$|z_n - z| \leq |a_n - a| + |b_n - b| < \varepsilon.$$

> **Correction note** In the original, only one direction of this equivalence was proved. The converse, which the teacher asked for, has been added. The definition of the limit must also be distinguished from the Cauchy criterion proper.

2.2 Limit theorems

Let $z_n \to z$ and $w_n \to w$. Then

$$z_n + w_n \longrightarrow z + w,$$
$$z_n - w_n \longrightarrow z - w,$$
$$z_n w_n \longrightarrow zw.$$

If $w \neq 0$, then w_n is also non-zero from some sufficiently large index onwards, and

$$\frac{z_n}{w_n} \longrightarrow \frac{z}{w}.$$

By the continuity of the modulus, it also follows that

$$|z_n| \longrightarrow |z|.$$

The statement about polar coordinates that followed in the original requires a restriction: if $z \neq 0$ and a continuous branch of the argument is chosen in a neighbourhood of z, then the correspondingly chosen arguments also converge. Without such a choice, angles are determined only up to multiples of 2π.

2.3 Example

Let

$$z_n = \frac{n^2 i^n}{n^3 + 1}.$$

Since $|i^n| = 1$,

$$|z_n| = \frac{n^2}{n^3 + 1} \leq \frac{1}{n} \longrightarrow 0.$$

Hence z_n converges to 0. For $\varepsilon > 0$ it suffices to choose some $N > 1/\varepsilon$, because then for all $n \geq N$

$$|z_n| < \varepsilon.$$

Geometrically, this means that from this index onwards all terms of the sequence lie inside the circle with centre 0 and radius ε.

> **Correction note** The original solution treated the numerator and denominator separately, even though their individual limits do not exist as finite numbers, and it reversed an ε-inequality incorrectly. The direct estimate of the modulus closes this gap without replacing the example.

3 The Mandelbrot set

3.1 General remarks

The Mandelbrot set, also known informally as the "Apfelmännchen" (literally "little apple man"), became widely known around 1980 through the work of Benoît B. Mandelbrot and is named after him. Mandelbrot worked at IBM's Thomas J. Watson Research Center in Yorktown Heights near New York. His book on the fractal geometry of nature made the new forms known beyond a small circle of specialists.

Mathematics increasingly concerned itself with complex feedback processes. Such processes have in principle long been known and, in simple cases, can be described by differential equations. Mandelbrot's work showed that iterated equations can also be used to investigate fragmented, irregular structures reminiscent of forms in nature, such as clouds or fern leaves. Computing the corresponding images became possible with fast computers.

An iteration uses the result of one computational step as the input of the next step. In general one writes

$$x_0 = k, \qquad x_{n+1} = f(x_n).$$

This gives

$$x_1 = f(k), \qquad x_2 = f(f(k)), \qquad x_3 = f(f(f(k))), \quad \ldots$$

The notation $f^2(x)$ here denotes the twofold application $f(f(x))$ and must not be confused with $(f(x))^2$. This procedure is called iteration or feedback.

3.2 Mathematical description of the Mandelbrot set

In this chapter we now attempt to present a complex function that opens up for us the world of aesthetic computer graphics. In principle, many complex functions can be iterated. For our example, the Apfelmännchen graphic, we choose the quadratic function

$$f_k(z) = z^2 + k, \qquad z = a + bi, \quad k = c + di, \quad a,b,c,d \in \mathbb{R}.$$

For the real and imaginary parts it follows that

$$f_k(a + bi) = (a^2 - b^2 + c) + i(2ab + d).$$

Every point of the complex plane is described by a complex number; its distance from the origin is the number's modulus. For a fixed constant k, the function is applied repeatedly, beginning with the starting value zero:

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + k.$$

The first iterations are therefore

$$z_1 = k, \qquad z_2 = k^2 + k, \qquad z_3 = (k^2 + k)^2 + k, \quad \ldots$$

For the image construction that follows, the decisive question is whether this sequence remains bounded or eventually escapes to infinity.

Correction note Despite the starting value $z_0 = 0$, the original records the expression $k^2 + k$ as the first step. It is in fact the second term of the sequence, z_2. The calculation has therefore only been re-indexed; the original approach is retained.

Three cases for the special case $k = 0$

The following case distinction from the original is correct for the special mapping $f_0(z) = z^2$, that is, for $k = 0$. Throughout, the moduli of the complex numbers are meant.

a) If $|z| < 1$, then $|z^{2^n}| = |z|^{2^n} \to 0$. The sequence approaches the origin.

b) If $|z| = 1$, all terms of the sequence have modulus 1. The terms themselves, however, need not be equal.

c) If $|z| > 1$, then $|z|^{2^n}$ grows without bound; the sequence escapes to infinity.

For the Mandelbrot set, however, it is $f_k(z) = z^2 + k$ that is investigated. Here the summand k must not be omitted. Membership therefore cannot be decided by the three cases at the unit circle alone. A simple counterexample is $k = -1$:

$$0 \longrightarrow -1 \longrightarrow 0 \longrightarrow -1 \longrightarrow \cdots$$

The orbit remains bounded, although it does not converge to zero.

Teacher's remark The red question "Wieso entfällt das k?" ("Why does the k disappear?") identifies precisely the gap in the original transition. The case distinction is therefore not deleted but explicitly retained as the special case $k = 0$.

3.3 The "Apfelmännchen" algorithm

Sequences that tend towards a fixed value or towards a periodic sequence are attracted by an attractor. If, on the other hand, the moduli escape without bound, one can speak intuitively of the attractor infinity. For the Mandelbrot set, however, it is not required that the sequence converge to zero. What matters is whether the orbit starting at zero remains bounded:

$$\mathcal{M} = \left\{ k \in \mathbb{C} \;\middle|\; z_0 = 0, \quad z_{n+1} = z_n^2 + k, \quad (z_n) \text{ is bounded} \right\}.$$

Calculating individual sequences by hand can show their behaviour only for a few examples. To investigate a larger region of the complex plane, the repeated calculation is transferred to the computer. Essentially three loops are needed for this: two loops run through the pixels, and the third carries out the iteration for each pixel.

Correction note Only the incorrect membership criterion has been replaced. The original defined the Mandelbrot set via the convergence $f^n(z) \to 0$. The correct criterion is the boundedness of the critical orbit; the point under investigation is the parameter k, not an arbitrary starting value.

Carrying out the algorithm

As an example, we choose the region $-2 \le c \le 1$ and $-1.5 \le d \le 1.5$ of the complex plane. Each pixel is assigned a constant $k = c + di$. During the calculation, the variables a and b hold

the real and imaginary parts of the current term of the sequence.

1. From the image size and the chosen region, determine the step sizes for c and d.
2. Assign the parameter $k = c + di$ to the current pixel.
3. Set $a := 0$, $b := 0$ and the iteration counter $h := N$.
4. Compute
$$\mathrm{Re} := a^2 - b^2 + c, \qquad \mathrm{Im} := 2ab + d.$$
5. Set $a := \mathrm{Re}$, $b := \mathrm{Im}$ and $h := h - 1$.
6. Repeat the calculation until $a^2 + b^2 > 4$ or $h = 0$.
7. If $h = 0$, the pixel is marked as not having escaped within the iteration limit. Otherwise it is certainly not a point of the Mandelbrot set.

The image region produced in this way is an approximation. With a finite number of iterations, a coloured point does not mean that membership has been proved, but only that the sequence has not escaped up to that point.

> **Correction note** In the original scheme the real and imaginary parts are computed but not assigned back to the variables a and b. Without this update no genuine iteration takes place. The two assignments have been added; the surviving Pascal listing also performs them.

Escape radius and Julia sets

Depending on the computing speed, the maximum number of iterations can be chosen, for example, between 25 and 150. A larger number improves the approximation, especially at the boundary, but increases the computing time.

For the critical orbit, the radius 2 suffices as the escape radius. If $|k| > 2$, then $z_1 = k$ already lies outside this circle. For $|k| \le 2$ and a later orbit point with $|z_n| > 2$ it follows that

$$|z_{n+1}| = |z_n^2 + k| \ge |z_n|^2 - |k| \ge |z_n|^2 - 2 > |z_n|.$$

The modulus then continues to grow and the orbit escapes to infinity. The equality $|z_n| = 2$ on its own, however, is not a uniform special case. For $k = -2$ the critical orbit reaches the value 2 and stays there; no general statement follows from this for all parameters or orbit points with modulus 2.

For comparison, we now consider the Julia sets. Mandelbrot and Julia sets arise from the same family of functions, but through two different questions:

- For the Mandelbrot set, the parameter k is varied while the starting value $z_0 = 0$ remains fixed.
- For a filled Julia set K_k, k is held fixed and the starting point z_0 is varied.

The two are linked by the important relationship

$$k \in \mathcal{M} \quad \Longleftrightarrow \quad K_k \text{ is connected.}$$

Here “connected” does not mean that any two points can necessarily be joined by a polygonal path within the set. That property would be stronger than the connectedness required here.

> **Historical context** The separation of the parameter plane and the dynamical plane, and the relationship between $k \in \mathcal{M}$ and the connectedness of K_k, were part of the theory in 1990. They are historical corrections and not modern standards applied after the fact.

3.4 Different ways of depicting Mandelbrot sets

Mandelbrot sets can be depicted in many ways. Depending on the chosen configuration, for instance a colour or a black-and-white display, different effects can be achieved. If the iteration process is stopped as soon as the modulus exceeds the given escape radius, the number of iterations needed up to that point can be assigned a colour or a grey level. This yields, among others, the following possibilities:

a) Only those points whose orbit does not escape within the iteration limit are marked.

b) Points with the same escape time receive the same colour. This produces coloured bands reminiscent of contour lines.

c) Combining the two methods gives a differentiated depiction with many gradations.

d) The escape speed can be interpreted as height. This produces a depiction that appears three-dimensional, for example a “Mandelbrot lake” with surrounding plateaus.

e) Assigning a different colour to each iteration step produces particularly colourful images.

f) Replacing colours with tones allows a musical composition to be created from the iteration data.

Many other forms of depiction, and variations on them, are also possible.

3.5 Similar structures within the Mandelbrot set

A further aspect directly connected with the Mandelbrot set is its approximate self-similarity or scale invariance. Objects are called self-similar if, when magnified, they show similar structures at different scales.

If one examines small regions at the boundary of the Mandelbrot set and magnifies them, ever new details appear. Miniatures of the Apfelmännchen shape occur that are related to the main figure. The Mandelbrot set, however, is not exactly self-similar everywhere; its repetitions are distorted and surrounded by new structures.

Such self-similar or approximately self-similar patterns are also found in nature, for instance in the branching of trees, ferns or river systems.

3.6 Further findings about Mandelbrot sets

The quadratic function investigated in this paper is the simplest case. Polynomials of higher degree can also be iterated and produce their own parameter and dynamical sets. For polynomials

of degree three, several critical points must in general be taken into account. Additional complex parameters can lead to higher-dimensional parameter spaces; a single complex variable, by contrast, remains representable in the two-dimensional complex plane.

Many depictions possess symmetries. Other structures show a sensitive transition between order and chaotic behaviour. Chaos theory investigates how very complicated behaviour can arise from simple deterministic rules.

Fractal geometry is not only a subject of theoretical research. Its basic ideas can also be conveyed in teaching and education, provided that a distinction is drawn between illustrative images, numerical approximations and mathematically proven statements.

4 Fractal dimension

4.1 The Koch snowflake

Classical Euclidean geometry essentially describes smooth shapes such as lines, circles, rectangles, spheres and pyramids. Many shapes in nature, however, are more complicated. A well-known example of a fractal curve is the Koch snowflake.

One starts with a line segment as the initiator. This is divided into three parts of equal length; the middle third is replaced by two sides of an equilateral triangle. The resulting polygonal line is called the generator. If each of its four segments is again replaced according to the same rule, and this process is continued indefinitely, the result is the Koch curve.

In the process the boundary becomes ever longer, while the resulting figure remains within a bounded region. The curve does not have the ordinary integer dimension of a line. It is more intricately branched than a line, yet it does not fill an area. For such fractured shapes Benoît Mandelbrot introduced the term “fractal”.

4.2 Fractal dimension

Suppose that each stage of an exactly self-similar figure consists of N reduced copies of the whole, each copy being scaled down by the factor $s > 1$. Then

$$N = s^D \qquad \Longrightarrow \qquad D = \frac{\log N}{\log s}.$$

This number D is called the similarity dimension. For a line segment, a square and a cube with the scaling factor $s = 3$, one obtains $N = 3$, $N = 9$ and $N = 27$, and hence the dimensions 1, 2 and 3.

For the Koch curve, each segment gives rise to four new segments, each scaled down by the factor 3. Therefore

$$D = \frac{\log 4}{\log 3} \approx 1.26186.$$

The dimension is thus not an integer, but lies between the dimension of a line and the dimension of a surface.

For sets that are not exactly self-similar, the formula above cannot be applied directly. A common numerical approximation is the box-counting method: one covers the set with squares of side length ε, counts the number $N(\varepsilon)$ of squares needed and examines the ratio

$$\frac{\log N(\varepsilon)}{\log(1/\varepsilon)} \quad \text{as} \quad \varepsilon \to 0.$$

> **Historical context** The original left it unclear whether the dimension referred to the whole Mandelbrot set or to its boundary. The Mandelbrot set itself has interior points and therefore has Hausdorff dimension 2. That its boundary also has Hausdorff dimension 2 was proved only after 1990 and must not be treated retrospectively as having been known at the time.

Further topics (2026)

5 Julia sets and connectedness

In Chapter 3, $z_{n+1} = z_n^2 + k$ was iterated with the fixed starting value $z_0 = 0$ (Sections 3.2 and 3.3). The question was for which parameters k the orbit remains bounded, that is, stays within a fixed circle about the origin (Chapter 2). Here the modulus $|z|$ is the distance from the origin (Chapter 1, Section 1.5).

If a point of the orbit lies outside the circle of radius 2, the critical orbit is certain to escape (Section 3.3). Shades of grey or colours show after how many steps a point escapes (Section 3.4). The Julia sets were mentioned only briefly at the end of Section 3.3.

This chapter reverses the question. We hold the parameter fixed and vary the starting value. Each parameter thus gives rise to its own image, the Julia set. In the end it turns out that the Mandelbrot set is a kind of table of contents of all these images.

On notation. In Chapter 3 the parameter was called $k = c + di$ with real numbers c and d. From now on, as in the specialist literature, we call the complex parameter itself c:

$$f_c(z) = z^2 + c, \qquad c \in \mathbb{C}.$$

The orbit of a starting value z_0 is z_0, $z_1 = f_c(z_0)$, $z_2 = f_c(z_1)$, $\ldots$

5.1 One fixed parameter, many starting values

For the simplest parameter $c = 0$ we have $f_0(z) = z^2$. Here the three cases from Section 3.2 give the complete answer: starting values with $|z_0| < 1$ run into the origin, starting values with $|z_0| > 1$ escape, and on the unit circle the modulus remains 1. The starting values with a bounded orbit therefore form the closed disc of radius 1; its boundary is the unit circle.

This point of view is now generalised. For a fixed parameter c,

$$K_c = \left\{ z_0 \in \mathbb{C} \;\middle|\; \text{the orbit } z_0, z_1, z_2, \ldots \text{ under } f_c \text{ is bounded} \right\}$$

is called the *filled Julia set* of f_c. Its boundary

$$J_c = \partial K_c$$

is called the *Julia set*. Every neighbourhood of a point of J_c, however small, therefore contains both starting values with a bounded orbit and escaping starting values [10]. For $c = 0$, K_0 is the disc and J_0 the unit circle.

All remaining points form the *Fatou set*. For quadratic functions it consists of the interior of K_c and the set of escaping starting values. On the Fatou set a small shift of the starting value changes the orbit only a little. One says that the iterates form a *normal family* there.

On the Julia set, by contrast, an arbitrarily small change of the starting value can decide whether the orbit remains bounded or escapes.

An escape criterion for arbitrary starting values

In Section 3.3 the radius 2 sufficed, because one always started at $z_0 = 0$. For arbitrary starting values the following holds: if

$$|z| > \max\bigl(2,|c|\bigr),$$

then the orbit of z escapes. The triangle inequality from Chapter 3 and $|c| < |z|$ give

$$|z^2 + c| \;\geq\; |z|^2 - |c| \;>\; |z|^2 - |z| \;=\; |z|\bigl(|z| - 1\bigr).$$

Since $|z| - 1 > 1$, we have $|z^2 + c| > |z|$; the new point of the orbit satisfies the condition again.

The modulus therefore grows in every step by at least the fixed factor $|z| - 1 > 1$ (with the starting value z), and the orbit escapes. In particular, K_c always lies in the disc of radius $\max(2,|c|)$. A bounded region of the plane is therefore sufficient for the images in this chapter.

Some simple properties

Because $f_c(-z) = f_c(z)$, z and $-z$ have the same orbit from the first step onwards. Every filled Julia set is therefore point-symmetric about the origin. Replacing c by $\bar{c}$ reflects the image in the real axis. For real c, K_c is therefore mirror-symmetric about the real axis and, together with the point symmetry, also about the imaginary axis.

If z lies in K_c, then $f_c(z)$ and the two preimages $\pm\sqrt{z - c}$ also lie in K_c, since their orbits agree apart from one step. K_c is therefore called *completely invariant*; the same holds for the boundary J_c.

The map f_c sends small pieces of J_c to larger ones, and the preimage branches send large pieces to small ones. Shapes therefore repeat on an ever smaller scale, but slightly distorted. Like the Mandelbrot set in Section 3.5, the Julia sets are only approximately self-similar.

> **Historical context** Gaston Julia [11] and Pierre Fatou [12, 13, 14] founded the theory of iterated rational functions independently of each other. Julia's memoir of 1918 received the Grand Prix of the Paris Académie des Sciences. There were no computer images; the sets were known only through their properties. The names "Julia set" and "Fatou set" became customary only later. In 1980 Mandelbrot published one of the first computer graphics of the set named after him [15]; the name "Mandelbrot set" was coined by Douady and Hubbard [16].

5.2 A gallery

An image of K_c is produced by the algorithm from Section 3.3, but with the roles exchanged: each pixel is a starting value z_0, and the parameter c is fixed for the whole image. Figure 1 shows eight examples.

1. $c = 0$, *circle*: K_0 is the disc, J_0 the unit circle.
2. $c = -1$, *basilica*: the orbit of 0 alternates between 0 and -1 (Section 3.2). K_{-1} consists of infinitely many black "bubbles" that touch at single points.
3. $c \approx -0.12256 + 0.74486i$, *Douady rabbit*: the orbit of 0 returns exactly to 0 after three steps. Bubbles meet three at a time in a point.

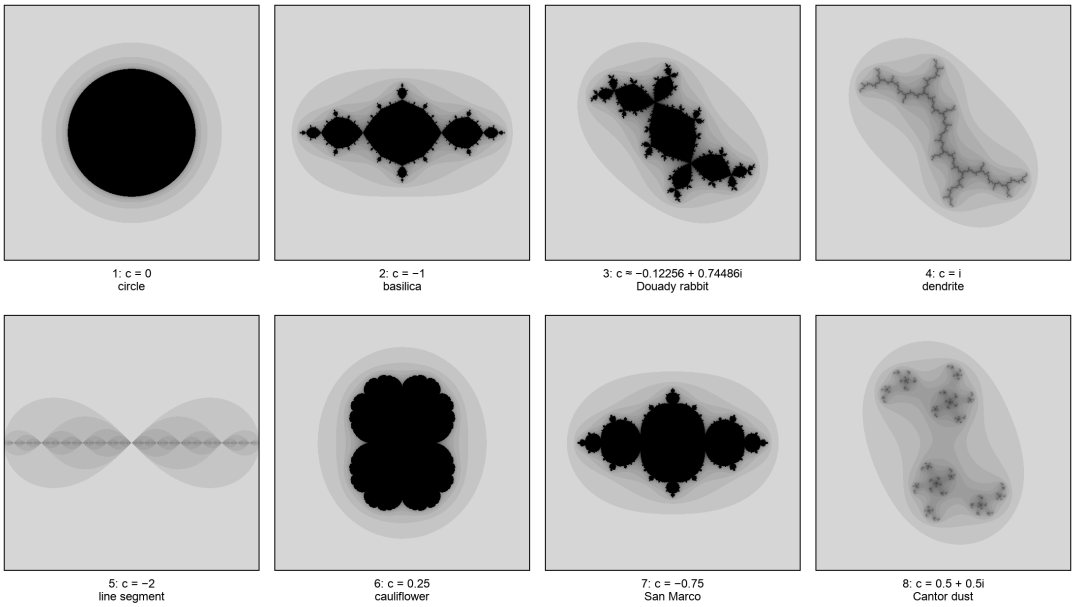


Figure 1: Eight filled Julia sets K_c in the dynamical plane (z-plane). Black: starting values whose orbit does not escape within the iteration limit. Shades of grey: escape time, light for fast, dark for slow escape. Images 1 to 7 belong to parameters in the Mandelbrot set and are connected; image 8 belongs to a parameter outside it and falls apart into dust.

4. $c = i$, *dendrite*: the orbit of 0 is $0 \to i \to -1 + i \to -i \to -1 + i \to \cdots$ and becomes periodic after a preperiod. K_i has no interior, so $K_i = J_i$. The set is finely branched like a tree.

5. $c = -2$, *line segment*: $K_{-2} = J_{-2}$ is the segment $[-2, 2]$ on the real axis.

6. $c = 0.25$, *cauliflower*: the orbit of 0 approaches the fixed point $\frac{1}{2}$ very slowly.

7. $c = -0.75$, *San Marco*: the name is Mandelbrot's. The set is a limiting form of the basilica.

8. $c = 0.5 + 0.5i$, *Cantor dust*: the orbit of 0 escapes after a few steps. K_c consists of uncountably many points, no two of which are connected to each other.

The names and descriptions follow [4, 10]. The images fall into two kinds: the first seven each hang together "in one piece", while the eighth has crumbled into dust. This observation leads to the most important result of the chapter.

5.3 What "connected" means

Intuitively, a set is *connected* if it consists of one piece. Mathematically this is formulated negatively: a set A is called connected if it cannot be split into two non-empty parts that are "separated" from each other by two disjoint open sets.

A stronger property is *path-connectedness*: any two points of A can be joined by a continuous curve that runs entirely within A. Every path-connected set is connected, but not conversely.

A well-known counterexample is the graph of $\sin(1/x)$ for $0 < x \leq 1$ together with the segment from $(0, -1)$ to $(0,1)$. The set is connected, yet no curve within it leads from the graph to

the segment. As in Section 3.3, what is meant in the following is always the weaker notion "connected".

5.4 The key: the orbit of the origin

The starting value 0 plays a special role, because the derivative $f_c'(z) = 2z$ vanishes only at $z = 0$. One calls 0 the *critical point* of f_c and its orbit the *critical orbit*. The Mandelbrot set consists exactly of the parameters with a bounded critical orbit. In the new language this reads

$$c \in \mathcal{M} \quad \Longleftrightarrow \quad 0 \in K_c.$$

The critical point decides the shape of the whole set K_c. For the quadratic functions the following holds [16, 10]:

$$\boxed{c \in \mathcal{M} \quad \Longleftrightarrow \quad 0 \in K_c \quad \Longleftrightarrow \quad K_c \text{ is connected}}$$

The same holds for J_c. This is why $\mathcal{M}$ is also called the *connectedness locus* of the quadratic family.

If c lies outside $\mathcal{M}$, then K_c even falls apart completely into dust. This dichotomy, its justification and the notion of the Cantor set are the subject of Chapter 6. The idea of the proof is this: as long as the critical point does not escape, the regions that approximate K_c from outside each remain a single piece; if it escapes, they are torn apart further and further.

A simple statement follows from this. Because $c = f_c(0)$, the orbit of c begins one step later than the orbit of 0. The one is bounded if and only if the other is. Hence we also have

$$c \in \mathcal{M} \quad \Longleftrightarrow \quad c \in K_c.$$

5.5 Two planes: dynamical plane and parameter plane

Both kinds of set are drawn in black and grey in the complex plane. Popular accounts therefore occasionally state: "The Julia set is a part of the Mandelbrot set." This statement is *false*.

The two sets lie in different planes. K_c and J_c consist of *starting values* z_0 and lie in the *dynamical plane* (z-plane); for each c there is a separate such plane. $\mathcal{M}$ consists of *parameters* c and lies in the single *parameter plane* (c-plane).

Both planes are copies of $\mathbb{C}$, so the images can be laid on top of each other. The same number, however, means a starting value in the one plane and a parameter in the other. Taken literally, the statement already fails for simple examples [10]:

- For $c = 0$, J_0 is the unit circle, so $z = 1$ lies on J_0. As a parameter, however, 1 does not belong to $\mathcal{M}$: the orbit $0 \to 1 \to 2 \to 5 \to 26 \to \cdots$ escapes.
- For $c = -2$, $J_{-2} = [-2, 2]$. On the real axis, however, $\mathcal{M}$ extends only from -2 to $\frac{1}{4}$. The points between $\frac{1}{4}$ and 2 lie in J_{-2} but not in $\mathcal{M}$.

What holds instead: $\mathcal{M}$ as a catalogue

What is correct is this: *each point* c of the parameter plane stands for *a whole* Julia set. Figure 2 shows the positions of the eight parameters of the gallery. $\mathcal{M}$ can therefore be read as a catalogue or table of contents of all quadratic Julia sets [17, 4]:

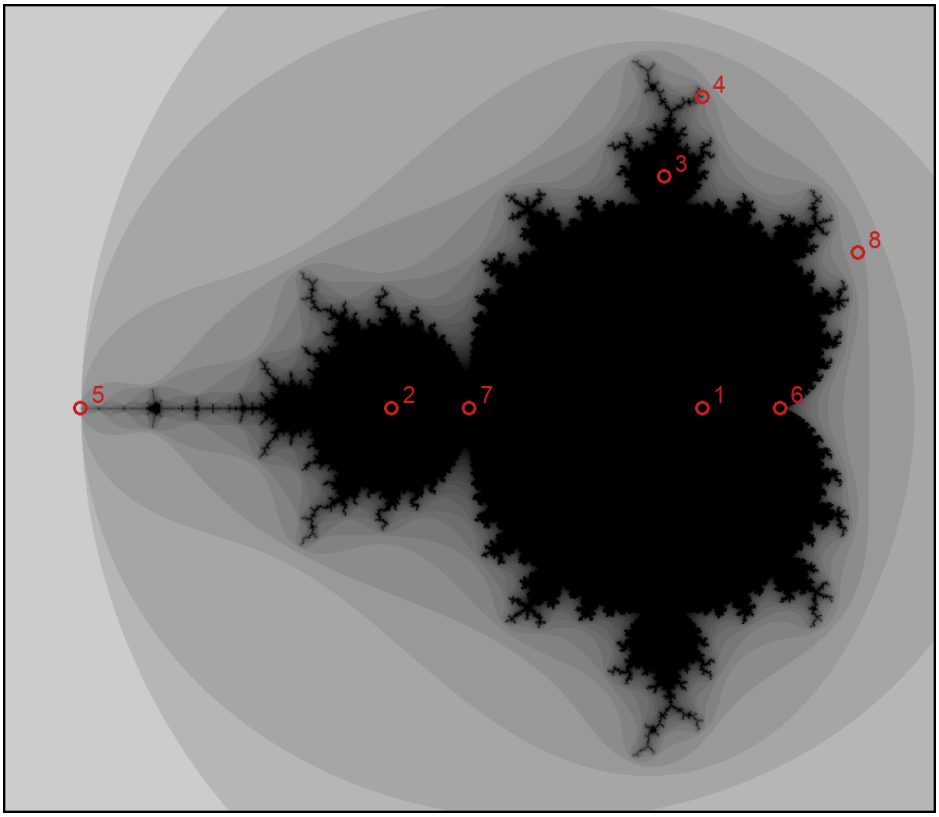


Figure 2: The Mandelbrot set $\mathcal{M}$ in the parameter plane (c-plane) with the eight parameters from Figure 1. Each marked point stands for a whole image in the dynamical plane. Points 1 to 7 lie in $\mathcal{M}$; point 8 lies outside.

- If c lies in $\mathcal{M}$, then J_c is connected (points 1 to 7).
- If c lies outside $\mathcal{M}$, then J_c is dust (point 8).

The position within $\mathcal{M}$ also says something about the image. Point 1 is the centre of the large main cardioid, point 2 the centre of the disc $|c+1| < \frac{1}{4}$ to its left. Point 3 lies at the centre of the "bud" on top of the cardioid. There the critical orbit is periodic with period 3; correspondingly, in the rabbit the bubbles meet three at a time.

Points 6 and 7 lie exactly on the boundary: at the cusp of the cardioid and at the point where cardioid and disc touch. Points 4 and 5 are tips of fine antennae; there the Julia set no longer has an interior.

Similarity at the boundary: Tan Lei's theorem

In one place the idea "Julia set in the Mandelbrot set" has a kernel of truth. We consider a parameter c_0 for which the critical orbit, after a preperiod, runs into a cycle without passing through 0 again. Examples are $c_0 = i$ with $0 \to i \to -1+i \to -i \to -1+i \to \cdots$ and $c_0 = -2$ with $0 \to -2 \to 2 \to 2 \to \cdots$. Such parameters are called *Misiurewicz points*; they lie on the boundary of $\mathcal{M}$.

In 1990 Tan Lei proved [18]: if one magnifies $\mathcal{M}$ about a Misiurewicz point c_0, and likewise J_{c_0} about the point $z = c_0$, ever more strongly, the two details become ever more similar to each other. In the limit they agree up to a rotation and scaling.

Magnifying $\mathcal{M}$ at the tip at point 4 therefore shows the same branchings as in the dendrite J_i at $z = i$. Only at such boundary points, and in the limit of ever stronger magnification, does one indeed find the Julia set "in" the Mandelbrot set.

This must be distinguished from the small Apfelmännchen that appear everywhere in $\mathcal{M}$ under magnification (Section 3.5). They are scaled-down copies of $\mathcal{M}$ itself, not Julia sets [19].

Exercises

1. Take $c = -1$ and the starting value $z_0 = 0.5$, and compute $z_1, z_2, \ldots, z_6$ with a calculator. As a check: $z_1 = -0.75$ and $z_2 = -0.4375$. Describe where the orbit seems to be heading and compare it with the orbit of 0. Repeat the calculation for $z_0 = 1.5$ and $z_0 = 1.7$, and decide which of these starting values lie in the basilica K_{-1}.

2. The following program checks whether a starting value z_0 escapes for a fixed c. It uses the escape criterion $|z| > \max(2, |c|)$ and outputs the number of steps until escape, or `None` if the orbit does not escape within `n_max` steps.

```
def escapes(z0, c, n_max=200):
    z = z0
    for n in range(n_max):
        if abs(z) > max(2, abs(c)):
            return n            # escaped after n steps
        z = z*z + c
    return None                 # did not escape

c = -1
for z0 in [0.5, 1.5, 1.7, 0.5+0.5j]:
    print(z0, escapes(z0, c))
```

```
print("Does 0 escape?", escapes(0, 0.5+0.5j))
```

Test further parameters from the gallery. `escapes(0, c)` checks (within the iteration limit) whether c belongs to the Mandelbrot set. Use `escapes(c, c)` to confirm that the orbits of 0 and c either both escape or both remain bounded.

3. Place Figure 1 next to Figure 2, read off the approximate position of c for each marker 1 to 8, and check whether it matches the stated value. Determine the marker that lies outside $\mathcal{M}$. For the parameter of point 1, $z = 1$ lies on the Julia set. Check whether the parameter $c = 1$ still lies within the region shown in Figure 2, and explain why it cannot lie in $\mathcal{M}$.

6 Mandelbrot, Julia and Cantor

In Chapter 5 we established that the filled Julia set K_c is connected if and only if c belongs to the Mandelbrot set $\mathcal{M}$ (Section 5.4); an orbit is bounded if it stays within a fixed circle about the origin (Chapter 2). This chapter deals with the case $c \notin \mathcal{M}$. Then K_c falls apart into infinitely many separate points; there is no intermediate state. The simplest example of such a "dust set" is the Cantor set.

6.1 The middle-thirds Cantor set

6.1.1 Construction

One begins with the interval $C_0 = [0, 1]$ and removes the open middle third $\left(\frac{1}{3}, \frac{2}{3}\right)$. Two intervals remain,

$$C_1 = \left[0, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, 1\right].$$

Removing the open middle third from each of them again gives four intervals of length $\frac{1}{9}$:

$$C_2 = \left[0, \tfrac{1}{9}\right] \cup \left[\tfrac{2}{9}, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, \tfrac{7}{9}\right] \cup \left[\tfrac{8}{9}, 1\right].$$

In general, C_n consists of 2^n intervals of length 3^{-n}. The *Cantor set* C is the set of points that are never removed, that is, the intersection

$$C = C_0 \cap C_1 \cap C_2 \cap \cdots = \bigcap_{n=0}^{\infty} C_n$$

[20]. Figure 3 shows the first stages.

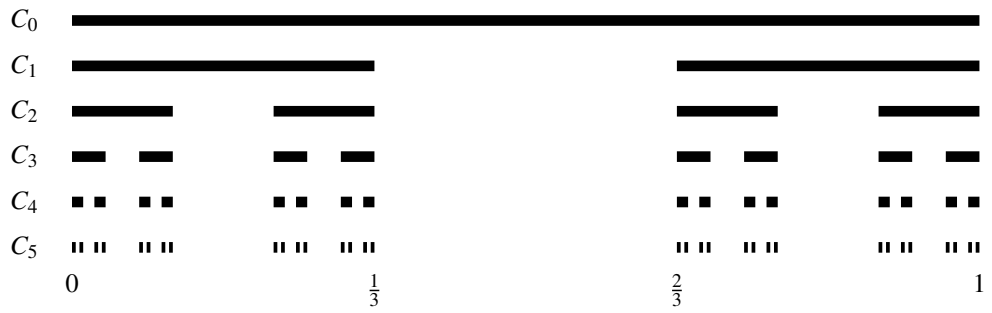


Figure 3: The first six stages $C_0, \ldots, C_5$ of the middle-thirds Cantor set. The open middle third is removed from every interval; 2^n intervals become 2^{n+1}.

Certainly all endpoints remain, for instance 0, $\frac{1}{3}$, $\frac{2}{9}$, $\frac{7}{9}$ or 1, because an endpoint is never removed. But far more remains.

6.1.2 The address in base three

Every number $x \in [0, 1]$ has a representation in base three (ternary representation)

$$x = 0.a_1a_2a_3\ldots_{(3)} = \frac{a_1}{3} + \frac{a_2}{9} + \frac{a_3}{27} + \cdots, \qquad a_k \in \{0,1,2\}.$$

The first digit indicates in which third of $[0,1]$ the number lies: $a_1 = 0$ for the left, $a_1 = 1$ for the middle, $a_1 = 2$ for the right third. The second digit determines the third within that third, and so on. Hence:

> x lies in C if and only if x has a representation in base three that uses only the digits 0 and 2.

The word "has" is important, because some numbers have two representations: $\frac{1}{3} = 0.1_{(3)} = 0.0222\ldots_{(3)}$. The second does without the digit 1, so the endpoint $\frac{1}{3}$ belongs to C.

The sequence of zeros and twos is called the *address* of the point. At each step it indicates the left or the right subinterval.

6.1.3 Properties

Uncountably many points. Every infinite sequence of zeros and twos yields a point of C, and different sequences yield different points. These sequences cannot be listed one by one.

Suppose $s^{(1)}, s^{(2)}, s^{(3)}, \ldots$ were a complete list of all such sequences. One forms a new sequence t whose k-th digit is a 2 if the k-th digit of $s^{(k)}$ is a 0, and vice versa. Then t differs from every sequence in the list in at least one place and is missing from the list, a contradiction.

By this *diagonal argument* of Cantor's, the Cantor set is *uncountable*, just like $[0,1]$. So it does not consist only of the countably many endpoints; $\frac{1}{4}$, for example, is not an endpoint and nevertheless lies in C (see the exercises at the end of the chapter).

Total length zero. In the first step an interval of length $\frac{1}{3}$ is removed, in the second two of length $\frac{1}{9}$, and in the $(n+1)$-th step 2^n intervals of length $3^{-(n+1)}$. The total length removed is a geometric series:

$$\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = 1.$$

Equivalently, C_n has total length $\left(\frac{2}{3}\right)^n \to 0$. Although C contains uncountably many points, it has length zero.

No interval, only dust. An interval of length $\ell > 0$ cannot lie in C, because it would have to lie in C_n, whose subintervals are shorter than ℓ for large n. Between any two points of C there is therefore a gap. Such a set is called *totally disconnected*: its only connected parts are single points.

Nevertheless, no point of C is isolated: changing a digit far back in the address gives further points of C arbitrarily close by. Closed sets without isolated points are called *perfect*. C is closed as an intersection of closed intervals; since C is also bounded, it is *compact*.

Self-similar. The parts of C in $\left[0, \frac{1}{3}\right]$ and in $\left[\frac{2}{3}, 1\right]$ are each copies of C reduced by the factor 3:

$$C = \tfrac{1}{3}C \;\cup\; \left(\tfrac{1}{3}C + \tfrac{2}{3}\right).$$

The left copy contains the addresses with first digit 0, the right one those with first digit 2. Like the Koch curve from Section 4.1, C is thus exactly self-similar, here with $N = 2$ copies and reduction factor $s = 3$. The formula from Section 4.2 gives

$$D = \frac{\log 2}{\log 3} \approx 0.6309.$$

The dimension lies between that of a point (0) and that of a line segment (1). In Chapter 7 we show that the Hausdorff dimension of the Cantor set also has this value.

> **Historical context** Georg Cantor published the middle-thirds set in 1883 in the fifth part of his work on infinite linear point sets, as an example of a perfect set that contains no interval [21]. In 1910 L. E. J. Brouwer proved that every non-empty, compact, perfect and totally disconnected set in the plane is topologically equivalent to the middle-thirds set, that is, can be mapped onto it continuously with a continuous inverse [22]. This is why all such sets are called *Cantor sets*.

6.2 The Fatou–Julia dichotomy

For $f_c(z) = z^2 + c$, 0 is the only *critical point*, that is, the only point with $f_c'(z) = 2z = 0$. Its orbit $0, c, c^2 + c, \ldots$ is called the *critical orbit*. By Section 3.3, c belongs to $\mathcal{M}$ if and only if this orbit remains bounded; as soon as a point of the orbit exceeds modulus 2, the orbit escapes. The following result goes back in essence to Pierre Fatou and Gaston Julia [12, 11].

Theorem (dichotomy).

(a) If the critical orbit is bounded, that is, $c \in \mathcal{M}$, then K_c and J_c are connected.

(b) If the critical orbit escapes, that is, $c \notin \mathcal{M}$, then $K_c = J_c$, and this set is a Cantor set: compact, perfect and totally disconnected.

[10, 23]

Dichotomy means division into two. In particular, for $z^2 + c$ there is no filled Julia set consisting of exactly two or exactly a hundred pieces. The Mandelbrot set thus sorts the Julia sets into two classes; hence the name *connectedness locus* from Section 5.4. In case (b) one speaks of *Cantor dust* or *Fatou dust*.

6.2.1 Idea of the proof

The proof rests on computing backwards. The preimages z of a point w, that is, the solutions of $z^2 + c = w$, are

$$z = \pm\sqrt{w - c}.$$

Every point $w \neq c$ therefore has two preimages, while the point $w = c$ has only one, namely $z = 0$. This explains the special role of the critical point.

Let V be a large, roundish region, a disc say, that contains K_c. For its preimage $f_c^{-1}(V)$ there are two cases.

- *The point c lies in V.* Then $V - c$ encloses the origin, and the two branches of the root $+\sqrt{\ }$ and $-\sqrt{\ }$ merge into each other. The preimage is again *one* connected region.
- *The point c does not lie in V.* Then the two branches of the root separate, and the preimage falls apart into *two* separate regions U_0 and U_1, each of which f_c maps onto the whole of V.

Repeated backward computation gives a sequence of nested regions with intersection K_c.

If the critical orbit is bounded, it lies in K_c and hence in each of these regions. The first case then always occurs, and the intersection K_c of connected regions is connected.

If the critical orbit escapes, V can be chosen so that it contains K_c but not the point c, and so that U_0 and U_1 again lie in V. The preimage then falls apart into two, four, eight, sixteen, ... parts, as in the Cantor construction. One also shows that the parts shrink down to single points.

Every point of J_c thereby receives an address of zeros and ones that states whether its orbit lies in U_0 or U_1 at each step; every such address belongs to exactly one point. The map f_c acts on these addresses by deleting the first digit [24, 25].

The same happens for the middle-thirds set: $x \mapsto 3x$ on $\left[0, \frac{1}{3}\right]$ and $x \mapsto 3x - 2$ on $\left[\frac{2}{3}, 1\right]$ delete the first digit of the ternary address. For $c \notin \mathcal{M}$ the dynamics of f_c on J_c is therefore essentially that on the Cantor set.

6.3 The real case $c < -2$

For real parameters $c < -2$ the disintegration can be followed on the number line. Since $\mathcal{M} \cap \mathbb{R} = \left[-2, \frac{1}{4}\right]$, these parameters lie outside $\mathcal{M}$ [10]. We calculate with $c = -2.5$, that is, $f(x) = x^2 - 2.5$.

The large interval. The larger fixed point of f is the solution of $x^2 + c = x$, namely

$$\beta = \frac{1 + \sqrt{1 - 4c}}{2} = \frac{1 + \sqrt{11}}{2} \approx 2.158.$$

For $|x| > \beta$ we have $f(x) > \beta$, and the orbit grows beyond all bounds. It therefore suffices to consider the interval $I = [-\beta, \beta]$.

Step 1: two intervals. The graph of f is an upward-opening parabola with vertex $(0, c)$. Since $c < -\beta$, the middle of I is mapped below $-\beta$ and escapes in the next step. More precisely, $f(x) < -\beta$ if and only if $x^2 < -\beta - c \approx 0.342$, that is, for $|x| < 0.585$. Two separate intervals remain,

$$I_0 \approx [-2.158, -0.585], \qquad I_1 \approx [0.585, 2.158].$$

Each of them is mapped by f strictly monotonically onto the whole of I, I_0 decreasingly and I_1 increasingly.

Step 2: four intervals. Since f maps each of I_0 and I_1 onto the whole of I, each of them contains an inner piece whose points fall into the gap $(-0.585, 0.585)$ in the next step. Four intervals remain,

$$[-2.158, -1.756], \quad [-1.384, -0.585], \quad [0.585, 1.384], \quad [1.756, 2.158].$$

Step n: 2^n intervals. After n steps there remain 2^n intervals whose points stay in I for at least n iterations (Figure 4). The points that stay in I for ever form the intersection of all stages, a Cantor set $\Lambda \subset \mathbb{R}$. All other real starting values escape.

All non-real starting values escape as well; hence $K_c = J_c = \Lambda$ [25]. That the intervals shrink down to points is harder to prove for c close to -2 than for strongly negative c.

The limiting case $c = -2$. For $c = -2$ we have $\beta = 2$, and the vertex lies exactly at $-2 = -\beta$. Nothing drops out any more: the intervals $[-2, 0]$ and $[0, 2]$ touch at 0, and $J_{-2} = K_{-2} = [-2, 2]$, the line segment from Section 5.2. An arbitrarily small step to the left of $c = -2$ is enough to make the segment fall apart into dust.

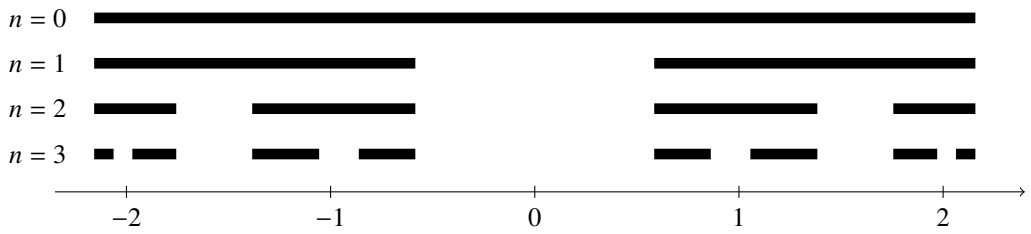


Figure 4: The real case $c = -2.5$: starting values whose orbit stays in the interval $I = [-\beta, \beta]$, $\beta \approx 2.158$, for at least 0, 1, 2 or 3 steps respectively. Unlike in the middle-thirds set, the gaps differ in size, because the parabola does not stretch equally strongly everywhere; the pattern "every interval falls apart into two" is the same.

The set Λ is a Cantor set in Brouwer's sense, but not the middle-thirds set. Its pieces are not reduced to scale but slightly distorted. Its dimension is in general not $\log 2/\log 3$ but depends on c (Chapter 7).

6.4 The transition in the plane

Figure 5 shows what happens when the parameter at the tip of the main body of $\mathcal{M}$, at $c = \frac{1}{4}$, is pushed to the right out of $\mathcal{M}$. For $c = 0.25$, K_c is the connected "cauliflower" from Section 5.2.

For $c = 0.26$ the critical orbit escapes, but exceeds modulus 2 only after 30 steps. By the theorem, $K_{0.26}$ is Cantor dust, even though the picture looks almost unchanged. The dark areas, including the black patches, do not belong to K_c; their starting values escape so slowly that they reach the iteration limit. At $c = 0.35$ the critical orbit escapes after 8 steps, and what remains of the cauliflower are scattered spirals of dust grains.

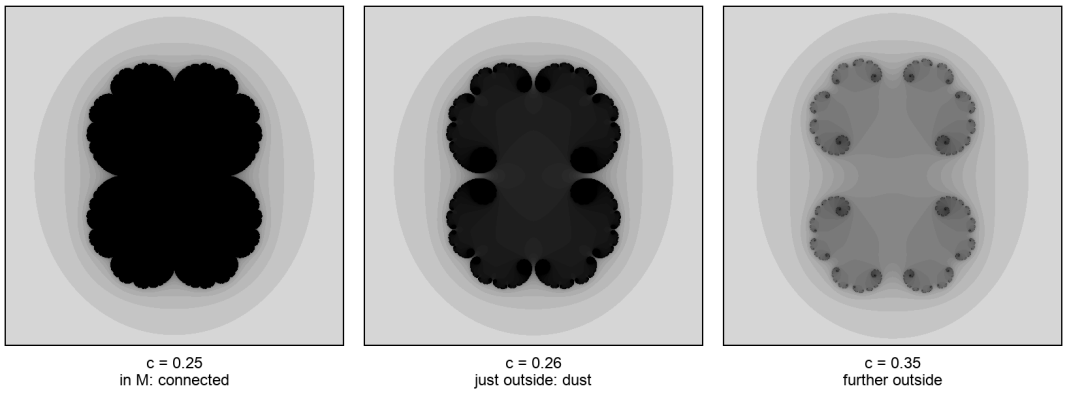


Figure 5: Filled Julia sets for $c = 0.25$ (in $\mathcal{M}$, connected), $c = 0.26$ (just outside $\mathcal{M}$) and $c = 0.35$ (further outside). Grey levels indicate the escape time: light means fast, dark slow escape, black no escape within the iteration limit. At $c = 0.26$, at finite resolution, the set looks almost connected; in fact it is totally disconnected.

This shows a limitation of all computer graphics: a pixel has finite size, and finitely many iterations cannot distinguish a very slow escape from remaining bounded. Whether a Julia set is connected is therefore decided with the theorem, by the *one* critical orbit.

6.5 Mandelbrot, Julia and Cantor: the connection

The results can be summarised as follows.

- Every point c of the parameter plane stands for a Julia set J_c in the dynamical plane. The Mandelbrot set is a kind of table of contents of these Julia sets.
- If c lies in $\mathcal{M}$, then J_c is connected: a circle, a basilica, a rabbit, a dendrite, a line segment.
- If c lies outside $\mathcal{M}$, then J_c is a Cantor set. All these sets are topologically equivalent to the middle-thirds set, but look very different and can have different dimensions.
- The decision is made by a single orbit, that of the critical point 0. It is precisely this orbit that the Apfelmännchen program from Section 3.3 computes.

The 1990 program therefore decided, approximately and for every pixel, whether the corresponding Julia set falls apart into dust. The "size" of such dust sets cannot be captured meaningfully by either a length or an area; how to measure it is the subject of Chapter 7.

Exercises

1. Compute the representations of $\frac{1}{4}$, $\frac{1}{3}$ and $\frac{1}{2}$ in base three by repeatedly multiplying by 3 and writing down the integer part as the next digit. The results are $\frac{1}{4} = 0.020202\ldots_{(3)}$, $\frac{1}{3} = 0.1_{(3)} = 0.0222\ldots_{(3)}$ and $\frac{1}{2} = 0.1111\ldots_{(3)}$. Determine which of the numbers lie in C and which of these is not an endpoint, and check the result for $\frac{1}{2}$ against Figure 3.

2. Compute the total length of $C_1, \ldots, C_5$ and the length removed after five steps. Confirm with the formula for the geometric series that in total the length 1 is removed. Determine after how many steps less than 1 % of the length remains.

3. For $c = -2.5$, the following program lays a fine grid of starting values on $[-2, 2]$ and counts how many still lie in $I = [-\beta, \beta]$ after 1,2,3, ... iterations.

```
c = -2.5
beta = (1 + (1 - 4*c) ** 0.5) / 2      # right fixed point, approx. 2.158
N = 100000                              # grid points on [-2, 2]
points = [-2 + 4*j/N for j in range(N + 1)]
for n in range(1, 11):
    points = [x*x + c for x in points]
    points = [x for x in points if abs(x) <= beta]
    print(n, len(points), round(len(points) / (N + 1), 4))
```

   After one step about 71 % remain, after ten steps around 6 %. Compute the ratio of successive counts and interpret the fact that it is roughly constant. Then replace c by -2.1 and by -2 and compare.

7 The Hausdorff dimension

Section 4.2 introduced the similarity dimension $D = \log N / \log s$ and box counting; for the Koch curve (Section 4.1) this gave $D = \log 4 / \log 3 \approx 1.26$. Chapter 6 constructed the Cantor set by removing middle thirds from $[0, 1]$ and, together with Chapter 5, showed that J_c is connected or a Cantor dust, depending on whether c lies in $\mathcal{M}$.

The similarity dimension applies only to figures made of exactly scaled-down copies of themselves, not, for instance, to the boundary of $\mathcal{M}$ or the path of a randomly wandering particle. This chapter introduces the Hausdorff dimension, which is defined for *all* sets. It agrees with the similarity dimension in the self-similar cases and allows statements such as "The boundary of $\mathcal{M}$ is two-dimensional."

7.1 The problem: length zero is not the same as length zero

After k construction steps of the Cantor set C, 2^k intervals of length 3^{-k} remain, first two of length $\frac{1}{3}$, then four of length $\frac{1}{9}$. The total length is

$$2^k \cdot 3^{-k} = \left(\tfrac{2}{3}\right)^k \longrightarrow 0 \qquad (k \to \infty).$$

The Cantor set therefore has length 0, like a single point, yet it contains uncountably many points (Chapter 6).

For such sets, length and area give only "zero" or "positive"; the Koch curve has infinite length and area zero. What is needed is a measure that resolves more finely and distinguishes a point from the Cantor set.

7.2 The idea: covering with small sets

One covers a set F with (countably many) pieces $U_1, U_2, \ldots$ of different sizes whose *diameter* $\operatorname{diam} U_i$, the greatest distance between two points, is at most δ. For an exponent $t \geq 0$ one forms

$$\sum_i (\operatorname{diam} U_i)^t .$$

We write t because s is the scaling factor in Section 4.2; in the specialist literature the exponent is usually called s. Taking the most economical covering and letting $\delta \to 0$ gives

$$\mathcal{H}^t_\delta(F) = \inf\Big\{ \sum_i (\operatorname{diam} U_i)^t \,:\, F \subseteq \bigcup_i U_i,\ \operatorname{diam} U_i \leq \delta \Big\}, \qquad \mathcal{H}^t(F) = \lim_{\delta \to 0} \mathcal{H}^t_\delta(F).$$

The infimum "inf" is, intuitively, the sum for the best covering. The limit exists because $\mathcal{H}^t_\delta(F)$ can only grow as δ becomes smaller. Without $\delta \to 0$ the Cantor set could be covered by one interval of length 1, and its fine structure would remain invisible.

The number $\mathcal{H}^t(F)$ is called the *t-dimensional Hausdorff measure* of F. For integer t: $\mathcal{H}^0$ counts points, $\mathcal{H}^1$ is length, and $\mathcal{H}^2$ is area up to a fixed factor [20].

Example 1 A point lies in a set of arbitrarily small diameter η. For $t > 0$, η^t is arbitrarily small, so $\mathcal{H}^t(\text{point}) = 0$; only for $t = 0$ does one obtain 1.

Example 2 A line segment of length 1 is divided into n sub-segments of length $\frac{1}{n}$. The sum is

$$n \cdot \left(\frac{1}{n}\right)^t = n^{1-t} \longrightarrow \begin{cases} \infty, & t < 1, \\ 1, & t = 1, \\ 0, & t > 1, \end{cases} \qquad (n \to \infty).$$

For $t > 1$ the segment is "too thin", for $t < 1$ "too large"; only $t = 1$ yields the length. That for $t < 1$ no cleverer covering keeps the sum small still has to be shown, but it is true.

7.3 The critical value

For every set there is exactly one value t_0 at which $\mathcal{H}^t(F)$ jumps from ∞ to 0. For $t < t_0$, $\mathcal{H}^t(F) = \infty$; for $t > t_0$, $\mathcal{H}^t(F) = 0$. Indeed, if $\operatorname{diam} U_i \le \delta$ and $t' > t$, then $(\operatorname{diam} U_i)^{t'} \le \delta^{t'-t}(\operatorname{diam} U_i)^t$. So if $\mathcal{H}^t(F)$ is finite, $\mathcal{H}^{t'}(F)$ is pushed down by the factor $\delta^{t'-t} \to 0$ and is zero.

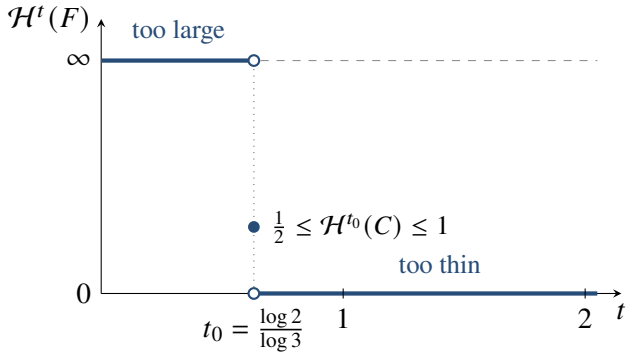


Figure 6: The t-dimensional Hausdorff measure of the Cantor set C as a function of t: infinite to the left of the critical value, zero to the right. Only at the critical value can a positive, finite number occur. This jump value is the Hausdorff dimension.

Definition. The jump value is called the *Hausdorff dimension* of F:

$$\dim_{\mathrm{H}} F = \inf\{t : \mathcal{H}^t(F) = 0\} = \sup\{t : \mathcal{H}^t(F) = \infty\}.$$

At the jump value itself, $\mathcal{H}^{t_0}(F)$ can be zero, positive and finite, or infinite; the dimension does not determine this.

The Cantor set. After k construction steps, C is covered by 2^k intervals of length 3^{-k} (Figure 7). The sum is

$$2^k \cdot \left(3^{-k}\right)^t = \left(\frac{2}{3^t}\right)^k.$$

For $3^t > 2$ it tends to 0; for $3^t < 2$ it grows without bound. The changeover lies at $3^t = 2$, that is, at

$$t_0 = \frac{\log 2}{\log 3} \approx 0.6309.$$

Since it uses one particular covering, the calculation shows only $\mathcal{H}^t(C) = 0$ for $t > t_0$. The harder part is that for $t < t_0$ no covering is substantially more economical: a small interval meets at most a few stage-k intervals of suitable size and can be replaced by these. It follows that $\frac{1}{2} \le \mathcal{H}^{t_0}(C) \le 1$ [20, Ch. 3], hence $\dim_{\mathrm{H}} C = \log 2/\log 3$.

A point therefore has dimension 0, the Cantor set dimension 0.63; with the same length 0, the Cantor set is measurably "larger".



Figure 7: The first stages of the Cantor set. Each stage is at the same time a covering of C: 2^k intervals of length 3^{-k}.

Countable sets have dimension 0. For a list $x_1, x_2, x_3, \ldots$ one covers x_k with an interval of length $\delta \cdot 2^{-k}$. For $t > 0$ the sum is at most $\delta^t\ (2^{-t} + 2^{-2t} + 2^{-3t} + \ldots)$, a geometric series times δ^t, and hence arbitrarily small. The rational numbers therefore have Hausdorff dimension 0, even though they lie densely; the uncountable Cantor set has positive dimension.

> **Historical context** Felix Hausdorff introduced these measures and fractional dimension in 1918 in the paper "Dimension und äußeres Maß" [26]; the volume of the *Mathematische Annalen* bears the date 1918/1919. His fully worked example was already the Cantor set with dimension $\log 2/\log 3$. Abram S. Besicovitch developed the theory from the late 1920s onwards [27]; hence the name Hausdorff–Besicovitch dimension. Mandelbrot's word "fractal" was added decades later [1].

7.4 Hausdorff dimension and similarity dimension

The calculation for the Cantor set was the formula $N = s^D$ from Section 4.2 with $N = 2$ and $s = 3$. Let a self-similar set F consist of m copies scaled down by the factors $r_1, \ldots, r_m < 1$. Its similarity dimension is the solution t of the *Moran equation*

$$r_1^t + r_2^t + \cdots + r_m^t = 1.$$

If all factors are equal to $r = 1/s$, it reads $m \cdot s^{-t} = 1$, so $m = s^t$, as in Section 4.2.

Unequal parts are also possible. If one keeps the pieces $[0, \frac{1}{2}]$ and $[\frac{3}{4}, 1]$ of $[0, 1]$ and repeats this, the equation reads $(\frac{1}{2})^t + (\frac{1}{4})^t = 1$. With $x = (\frac{1}{2})^t$ it follows that $x + x^2 = 1$, so $x = \frac{\sqrt{5}-1}{2}$ and $t = \log \frac{1+\sqrt{5}}{2} \,/\, \log 2 \approx 0.694$.

Theorem (Moran 1946, Hutchinson 1981). If the copies do not overlap substantially, then the Hausdorff dimension and the box dimension are equal to the similarity dimension [28, 29]; see also [20, Sect. 9.2].

"Not overlapping substantially" is made precise by the *open set condition*: there is a non-empty open set V whose scaled-down copies $S_1(V), \ldots, S_m(V)$ lie in V and are pairwise disjoint. Contact at the boundary, as between the parts of the Koch curve or at the corners of the Sierpiński triangle, is allowed. For all the examples from Section 4.2, the formula $D = \log N/\log s$ therefore gives the Hausdorff dimension.

When the copies overlap. The segment $[0, 1]$ is the union of the three half-length segments $[0, \frac{1}{2}]$, $[\frac{1}{4}, \frac{3}{4}]$ and $[\frac{1}{2}, 1]$. Here the formula gives $\log 3/\log 2 \approx 1.58$, more than the dimension 1 of the segment, because the middle copy lies almost entirely on the others. With overlaps, the

dimension can therefore be smaller than the similarity dimension; when exactly is a current research question (Chapter 9).

7.5 Box dimension and Hausdorff dimension

In the box counting of Section 4.2, all pieces are of the *same* size: one counts the smallest number $N_\delta(F)$ of sets of diameter at most δ that cover F, and examines $\log N_\delta(F)/\log(1/\delta)$ as $\delta \to 0$. Since the Hausdorff dimension allows pieces of different sizes, it is always the case that

$$\dim_{\mathrm{H}} F \;\le\; \dim_{\mathrm{Box}} F,$$

provided the limit defining the box dimension exists; otherwise one takes the lower limit. Equality holds for self-similar sets satisfying the open set condition, but not in general.

Example. The set $F = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$ is countable, so $\dim_{\mathrm{H}} F = 0$. The distance between $\frac{1}{n}$ and $\frac{1}{n+1}$ is $\frac{1}{n(n+1)} \approx \frac{1}{n^2}$. For $n < 1/\sqrt{\varepsilon}$ it is greater than ε, and each of these points needs a box of width ε of its own: about $1/\sqrt{\varepsilon}$ of them. The remaining points fill $[0, \sqrt{\varepsilon}\,]$ so densely that the whole interval has to be covered; that is another roughly $\sqrt{\varepsilon}/\varepsilon = 1/\sqrt{\varepsilon}$ boxes. Hence $N(\varepsilon) \approx 2\,\varepsilon^{-1/2}$ and

$$\dim_{\mathrm{Box}} F = \frac{1}{2} \neq 0 = \dim_{\mathrm{H}} F \qquad [20, \text{Ch. 2}].$$

The rational numbers in $[0, 1]$ lie densely, so every covering by boxes covers the whole interval: box dimension 1, Hausdorff dimension 0.

Box counting is easy to compute: a computer counts the occupied boxes of a grid and fits a straight line in a log–log plot. The Hausdorff dimension is the mathematically better quantity; for instance, the dimension of a countable union is the supremum of the individual dimensions, which does not hold for the box dimension.

7.6 A table of values

The columns for length and area in Table 1 follow the jump rule from Section 7.3: for dimension below 1 the length is zero, above it infinite; the same holds for area with the threshold 2.

The Brownian path is a *curve* and yet has dimension 2, but no area. The area of $\partial\mathcal{M}$ is unknown; this is the subject of the next section.

7.7 Dimension in complex dynamics

Julia sets for small c. For $c = 0$, J_0 is the unit circle (Chapter 5), with dimension 1. In the main cardioid of $\mathcal{M}$, J_c is a crumpled closed curve without self-contact. Ruelle proved in 1982 that $\dim_{\mathrm{H}} J_c$ depends very smoothly (even real-analytically) on c there [32]. In the specialist literature the approximation

$$\dim_{\mathrm{H}} J_c \approx 1 + \frac{|c|^2}{4 \ln 2}$$

is given for small $|c|$ and is usually attributed to Ruelle; whether it appears in this form in his original paper is unclear. For $c = 0.1$ it would give about 1.0036. Even a small perturbation of z^2 thus turns the circle into a fractal.

Set	Structure	$\dim_H$	Length $\mathcal{H}^1$	Area
Point	–	0	0	0
Line segment of length 1	3 copies, factor $\frac{1}{3}$	1	1	0
Cantor set (Ch. 6)	2 copies, factor $\frac{1}{3}$	$\frac{\log 2}{\log 3} \approx 0.6309$	0	0
Koch curve (Sect. 4.1)	4 copies, factor $\frac{1}{3}$	$\frac{\log 4}{\log 3} \approx 1.2619$	∞	0
Sierpiński triangle	3 copies, factor $\frac{1}{2}$	$\frac{\log 3}{\log 2} \approx 1.5850$	∞	0
Square	9 copies, factor $\frac{1}{3}$	2	∞	positive
Path of a Brownian motion in the plane	random	2 (almost surely) [30]	∞	0
Boundary of the Mandelbrot set $\partial\mathcal{M}$	no exact self-similarity	2 [31]	∞	unknown

Table 1: Hausdorff dimension, length and area of some sets. The first six values follow from the Moran equation with the open set condition. The Brownian path is the trace of a randomly wandering particle; that its area is zero goes back to Paul Lévy [20, Ch. 16].

Cantor dust. If c lies outside $\mathcal{M}$, then J_c is a Cantor dust (Chapter 6). Its dimension tends to 0 as $|c|$ grows.

The boundary of the Mandelbrot set. The Mandelbrot set has interior points, for example the main cardioid, and therefore trivially has dimension 2. Its *boundary* $\partial\mathcal{M}$, on the other hand, contains no disc. One would expect a dimension slightly above 1, as for the boundary of the Koch snowflake with 1.26.

Theorem (Shishikura). $\dim_H \partial\mathcal{M} = 2$. Moreover, $\dim_H J_c = 2$ for "typical" parameters c on $\partial\mathcal{M}$, more precisely for a set of boundary points that is large in the topological sense, a so-called residual set [31].

In the sense of dimension, the boundary is therefore as large as a piece of the plane, without "filling" any area; more is not possible in the plane. By the jump rule, $\mathcal{H}^t(\partial\mathcal{M}) = \infty$ for every $t < 2$; in particular, the length is infinite. The magnifications from Section 3.5 give an impression of this.

The proof starts from parameters at which a periodic point becomes "parabolic", as at $c = \frac{1}{4}$ at the cusp of the main cardioid. Small shifts of c produce subsets of the Julia sets with dimension arbitrarily close to 2. This fine structure is transferred to the parameter plane, that is, to $\partial\mathcal{M}$; this fits with $\mathcal{M}$ as a "catalogue" of the Julia sets (Chapter 5).

Dimension 2 does *not* mean positive area, as the Brownian path shows. Whether $\partial\mathcal{M}$ has positive area is open (Chapter 9). Dimension 2 also occurs for Julia sets of other functions, for example for λe^z [33] (Chapter 8).

> **Historical context** Shishikura's result circulated as a preprint from 1991; a short version appeared in 1994, and the complete paper in 1998 in the *Annals of Mathematics*, that is, after the original 1990 version of this paper. It settles the question left open there (Section 4.2): the boundary of $\mathcal{M}$, too, has dimension 2.

7.8 Significance of the Hausdorff dimension

Size beyond length and area. The Cantor set, the rational numbers, the Koch curve and the Brownian path all have area 0. Their dimensions 0.63, 0, 1.26 and 2 nevertheless clearly distinguish them.

Measuring exceptional sets. Many theorems hold "almost everywhere", that is, except on a null set; the dimension measures these exceptions. The numbers in $[0,1]$ in whose decimal expansion the digit 7 never occurs form a set of length 0 made of 9 copies with the factor $\frac{1}{10}$, and so of dimension $\log 9/\log 10 \approx 0.954$. The Cantor set is the same thing for base three and the digit 1 [20, Ch. 10].

Scaling laws in nature and technology. Lewis F. Richardson found that the length of coastlines measured with rulers of length δ grows like δ^{1-D}, for the west coast of Britain with $D \approx 1.25$ [34]. In 1967 Mandelbrot interpreted D as a fractional dimension [35]. Similar power laws occur in growth processes, turbulence and porous materials; in medicine, blood vessels or bronchi are described in this way. Real objects, however, are not mathematical sets, and measured data span only a few orders of magnitude. A D obtained in this way is an empirical box-counting exponent with model assumptions, not a proven value of a Hausdorff dimension [20].

A solved problem. For the outer boundary of the trace of a random walker in the plane, Mandelbrot conjectured the dimension $\frac{4}{3}$ on the basis of computer images. Lawler, Schramm and Werner proved this in 2001 [36]; more on this in Chapter 9.

Exercises

1. Draw the Koch curve up to the third stage on squared paper with a base segment of 27 squares, lay grids with box width 9, 3 and 1, that is, $\varepsilon = \frac{1}{3}, \frac{1}{9}, \frac{1}{27}$ of the base segment, over it and in each case count the boxes that the curve touches. Determine the factor by which the number grows when ε shrinks to a third, compare $\log(\text{factor})/\log 3$ with $\log 4/\log 3 \approx 1.26$ and explain the discrepancy.

2. A square is divided into 3×3 equal sub-squares; only the four corner squares are kept, and the procedure is repeated in each remaining square ("Cantor dust in the square"). Compute the dimension with $N = s^D$. The result $\log 4/\log 3$ is the same as for the connected Koch curve; explain what the dimension does *not* say about a set. Variant: also keep the centre square (5 of 9), or keep the pieces $[0, \frac{1}{2}]$ and $[\frac{3}{4}, 1]$ of $[0,1]$ and solve the Moran equation from Section 7.4.

3. The following program determines the box dimension of the Cantor set. It generates the 2^{15} left endpoints of the stage-15 intervals and counts the occupied boxes of width 2^{-m}. It prints m, N, the quotient $\log N/\log(1/\varepsilon)$ and the slope in the log–log plot between ε and $\varepsilon/16$.

```
from math import log
pts = [0.0]                       # left ends of the Cantor intervals
for k in range(15):
    pts = [x / 3 for x in pts] + [x / 3 + 2 / 3 for x in pts]
N = {}
for m in range(2, 19):
    eps = 2 ** -m                 # box width 1/2^m
    N[m] = len({int(x / eps) for x in pts})    # occupied boxes
```

```
    q = log(N[m]) / log(1 / eps)
    if m >= 6:
        s = log(N[m] / N[m - 4]) / log(16)     # slope in the log-log plot
        print(m, N[m], round(q, 3), round(s, 3))
```

For $m = 18$ the quotient gives only 0.683, whereas the slope gives 0.625, close to the exact value 0.6309. Explain why the quotient falls so slowly (hint: $N \approx K \cdot \varepsilon^{-D}$ with a constant factor K) and why the slope fluctuates (hint: the base-two grid does not match the base-three structure of the set). Also examine box widths 3^{-m}.

8 Beyond $z^2 + c$

Chapters 5 and 6 studied the family $f_c(z) = z^2 + c$: K_c is connected if and only if the orbit of 0 remains bounded; otherwise J_c is a Cantor dust. An escape radius of 2 suffices (Section 3.3). The 1990 Facharbeit already hinted at the end of Chapter 3 that other functions can be iterated as well.

This chapter follows that path: to higher powers, to general polynomials and rational functions, and finally to the transcendental functions e^z, $\sin z$, $\cos z$, $\tan z$ and $\ln z$. The tools are the polar form $z = r(\cos\varphi + i\sin\varphi)$ from Section 1.6.2 and the Hausdorff dimension from Chapter 7, which also measures sets without length or area.

8.1 Higher powers: $z^d + c$ and Multibrot sets

For a natural number $d \geq 2$ we consider

$$f_{d,c}(z) = z^d + c.$$

The derivative $d\,z^{d-1}$ vanishes only at $z = 0$. As for $z^2 + c$, there is therefore exactly one critical point, and the dynamics depend on its orbit 0, c, $c^d + c$, $\ldots$

The dichotomy from Chapter 6 holds unchanged: if the orbit of 0 remains bounded, the filled Julia set is connected; otherwise it is a Cantor dust [10]. The set

$$\mathcal{M}_d = \{\, c \in \mathbb{C} \mid \text{the orbit of 0 under } f_{d,c} \text{ is bounded} \,\}$$

is called the *Multibrot set* of degree d; $\mathcal{M}_2 = \mathcal{M}$. The escape radius 2 remains usable: the estimate from Section 3.3 holds word for word, because $|z|^d \geq |z|^2$ for $|z| > 2$.

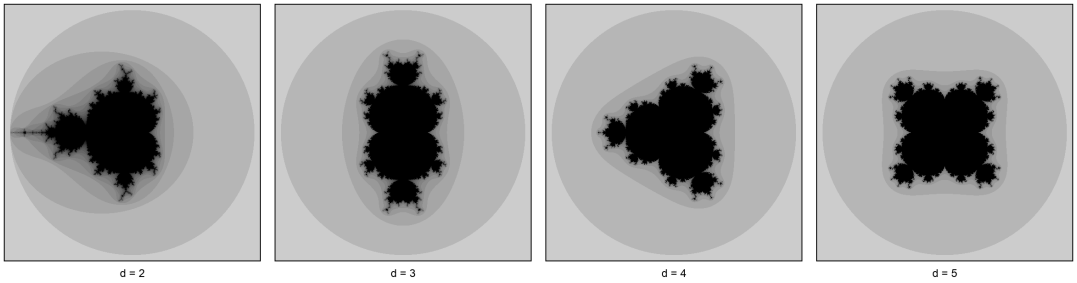


Figure 8: The Multibrot sets $\mathcal{M}_d$ for d = 2,3,4,5 in the parameter plane. Black: the orbit of 0 has not escaped within the iteration limit. Shades of grey: escape time as in Section 3.4. $\mathcal{M}_3$ is point-symmetric (twofold), $\mathcal{M}_4$ has threefold and $\mathcal{M}_5$ fourfold symmetry.

Rotational symmetry: $d - 1$ instead of d

According to Figure 8, $\mathcal{M}_d$ has $(d-1)$-fold rotational symmetry, whereas the associated Julia sets have d-fold symmetry. Both follow from the polar form. Let $\omega = \cos\frac{2\pi}{m} + i\sin\frac{2\pi}{m}$. By de Moivre, $\omega^m = 1$, and multiplication by ω rotates the plane by $360°/m$ about the origin.

Julia sets ($m = d$). From $\omega^d = 1$ it follows that $f_{d,c}(\omega z) = \omega^d z^d + c = f_{d,c}(z)$. The points z and ωz therefore have the same image after one step; either both remain bounded or both escape.

The filled Julia set is therefore invariant under rotation by $360°/d$. For $d = 2$ this is the point symmetry $z \mapsto -z$ from Chapter 5.

Multibrot sets ($m = d - 1$). Now let $\omega^{d-1} = 1$, so $\omega^d = \omega$. We compare the critical orbit $z_0 = 0$, $z_{n+1} = z_n^d + c$ with the orbit $w_0 = 0$, $w_{n+1} = w_n^d + \omega c$ for the rotated parameter ωc. From $w_n = \omega z_n$ (clear for $n = 0$) it follows that

$$w_{n+1} = (\omega z_n)^d + \omega c = \omega^d z_n^d + \omega c = \omega\,(z_n^d + c) = \omega z_{n+1}.$$

The orbit for the parameter ωc is therefore the orbit for the parameter c rotated by $360°/(d-1)$, with the same moduli. Hence $c \in \mathcal{M}_d \Leftrightarrow \omega c \in \mathcal{M}_d$.

One factor ω is "used up" in turning ω^d back into ω; hence the condition $\omega^{d-1} = 1$. For $d = 2$, $\omega = 1$: the Mandelbrot set has no rotational symmetry, only the reflection symmetry in the real axis. This holds for all $\mathcal{M}_d$ because $\overline{z^d + c} = \overline{z}^d + \overline{c}$.

8.2 General polynomials and rational functions

Several critical points. A polynomial of degree d has, counted with multiplicity, $d - 1$ critical points, the zeros of its derivative. For $z^d + c$ they all coincide at 0; in general they have different orbits. Already for degree 3 one therefore needs two complex parameters, for instance in the normal form

$$P_{a,b}(z) = z^3 - 3a^2 z + b$$

with the critical points $\pm a$.

The set of all (a,b) for which both critical orbits remain bounded is the cubic counterpart of $\mathcal{M}$. It lies in $\mathbb{C}^2$, so it has four real dimensions and can only be drawn in slices. By Branner and Hubbard, it is compact and connected [37]. This is the precise form of the cautious remark in the original about higher-dimensional parameter spaces (Chapter 3).

Limits of the dichotomy. For all polynomials of degree $d \geq 2$ only a weakened dichotomy holds [10]: if *all* critical orbits remain bounded, the filled Julia set is connected; if *at least one* escapes, it falls apart into uncountably many pieces. For $z^2 + c$, in the second case every piece is a single point.

For cubic polynomials, however, one critical orbit can escape while the other remains bounded. Then some pieces can be shrunken, distorted copies of a connected quadratic Julia set; the Julia set is neither connected nor a Cantor dust [38]. Exactly which polynomials have Cantor sets as Julia sets is described by a conjecture of Branner and Hubbard that was proved in 2009 [39].

That a quadratic Julia set can be "hidden" inside a cubic polynomial is explained by Douady and Hubbard's theory of polynomial-like mappings [19]. It also accounts for the small Apfelmännchen from Section 3.5.

Rational functions and Newton's method. Julia and Fatou developed their theory from the outset for rational functions $R(z) = P(z)/Q(z)$. An example from the school syllabus is Newton's method: for a polynomial p one iterates the rational function

$$N_p(z) = z - \frac{p(z)}{p'(z)}.$$

Every simple zero of p is a strongly attracting fixed point of N_p.

For $p(z) = z^3 - 1$ there are three zeros, 1 and $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, and hence three basins of attraction. Their common boundary is the Julia set of N_p, a fractal: at each of its points all three basins meet. Near this boundary a tiny change in the starting value decides which zero one finds [4]. For rational functions the Fatou components can be completely classified. The decisive step was Sullivan's proof in 1985 that there are no "wandering" components [40].

8.3 The bridge to transcendental functions

Euler's formula from the polar form. By Section 1.6.2, the addition formulae give, for $E(\varphi) = \cos\varphi + i\sin\varphi$,

$$E(\varphi_1)\,E(\varphi_2) = E(\varphi_1 + \varphi_2), \qquad E(0) = 1.$$

This is the same rule as for powers, $a^x a^y = a^{x+y}$. The derivative fits too: $E'(\varphi) = -\sin\varphi + i\cos\varphi = i\,E(\varphi)$. So E behaves as $e^{i\varphi}$ would have to behave if the rule $(e^{k\varphi})' = k\,e^{k\varphi}$ is also to hold for $k = i$. One therefore defines

$$e^{i\varphi} = \cos\varphi + i\sin\varphi \qquad \text{(Euler's formula)}.$$

The polar form thus becomes $z = r\,e^{i\varphi}$, and the rule "multiply the moduli, add the angles" is the law of exponents. For $\varphi = \pi$ one obtains $e^{i\pi} = -1$. The usual proof substitutes $i\varphi$ into the power series of the exponential function.

The complex exponential function. For $z = x + iy$ one sets

$$e^z = e^x e^{iy} = e^x(\cos y + i\sin y).$$

So $|e^z| = e^x$ and $\arg e^z = y$: the real part determines the modulus, the imaginary part the angle. In particular $e^z \neq 0$ and $e^{z+2\pi i} = e^z$.

The exponential function therefore has the purely imaginary period $2\pi i$ and takes every value $w \neq 0$ infinitely often. Its images repeat in the vertical direction every $2\pi \approx 6.28$ units. From e^{iz} and e^{-iz} one obtains

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

For real z these are the familiar functions with period 2π. In the complex plane they are unbounded: for example, $\cos(i) = \frac{e^{-1}+e}{2} \approx 1.543$, and $|\sin(x+iy)|$ grows roughly like $\frac{1}{2}e^{|y|}$.

Escape without an escape radius. For a polynomial of degree $d \geq 2$, $|P(z)|$ is much larger still when $|z|$ is large; the escape radius 2 of the 1990 algorithm rests on this. For e^z this does not hold: $z = -1000$ has a large modulus, but e^{-1000} lies very close to 0. One says that e^z has an *essential singularity* at ∞: outside every circle, however large, e^z takes every value except 0 infinitely often. There is therefore no escape radius. The same holds for $\sin z$ and $\cos z$; for example, they remain bounded on the whole real axis.

Moreover, for λe^z, $\lambda\sin z$ and $\lambda\cos z$, all points whose orbit tends to ∞ lie in the *Julia set* [41]. For $z^2 + c$ the escaping points formed an open region of the Fatou set outside K_c. In the following images, by contrast, they show the Julia set itself.

8.4 The exponential family λe^z

We iterate $E_\lambda(z) = \lambda e^z$ with $\lambda \neq 0$. Since $E'_\lambda = E_\lambda$ is never zero, there is no critical point. Its role is taken over by the value 0, which E_λ never takes but which the function approaches arbitrarily closely in the left half-plane. As for $z^2 + c$, then, the orbit of a single "special" value, $0, \lambda, \lambda e^\lambda, \ldots$, governs the dynamics [42].

Two fixed points and their merging. For real $\lambda > 0$ the graphs of $y = \lambda e^x$ and $y = x$ intersect twice for small λ and not at all for large λ. In the limiting case they touch: $\lambda e^x = x$ and at the same time $\lambda e^x = 1$ (equal gradient). It follows that $x = 1$ and $\lambda = 1/e \approx 0.3679$.

For $0 < \lambda < 1/e$ the smaller fixed point q is attracting, since $E'_\lambda(q) = \lambda e^q = q < 1$; for $\lambda = 0.3$, $q \approx 0.4894$. The orbit of 0 tends to q, and the Fatou set is precisely the basin of attraction of q [43].

The Cantor bouquet. The Julia set then consists of uncountably many pairwise disjoint curves that run off to infinity to the right, each with one endpoint. They are called *hairs*, and the whole is called a *Cantor bouquet* [44, 45]. Each hair is approximated arbitrarily closely from both sides by other hairs, much as every point of the Cantor set is a limit of other points. On the hairs the orbits flee to the right; the points in between are drawn towards q.

The Julia set has Hausdorff dimension 2 [33], but area 0. The hairs without their endpoints have dimension only 1, as is to be expected of curves. The full dimension therefore resides in the set of endpoints [46, 47].

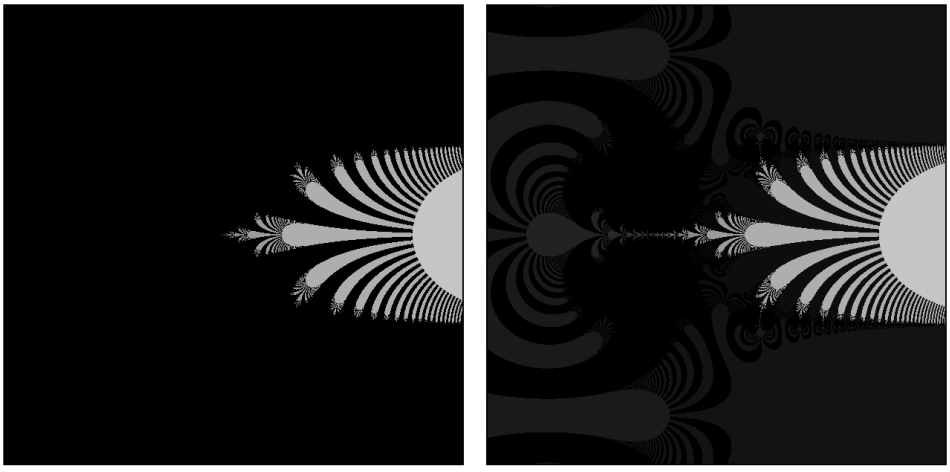


Figure 9: The exponential family λe^z for $\lambda = 0.3$ (left) and $\lambda = 0.4$ (right). *The shades of grey are reversed compared with the images of* $z^2 + c$. Light: the orbit flees to infinity to the right; these points lie on the hairs and belong to the Julia set, not to the Fatou set. On the left, black is the basin of attraction of the attracting fixed point $q \approx 0.49$, that is, the Fatou set. On the right there is no attracting fixed point; the whole plane is Julia set, and dark points have merely not escaped within the iteration limit.

The explosion at $\lambda = 1/e$. At $\lambda = 1/e$ the two fixed points merge at the point 1. For real $\lambda > 1/e$ there is no longer a real fixed point, the orbit of 0 travels along the real axis to infinity, and the Julia set is the whole plane [48]. The step from $\lambda = 0.36$ to $\lambda = 0.37$ thus turns a

nowhere dense set into the whole of $\mathbb{C}$. This is called an *explosion* of the Julia set. For $z^2 + c$ this is impossible, because there the Julia set is always bounded and has no interior.

For $\lambda = 1$, that is, for e^z itself, the orbit $0 \to 1 \to e \to e^e \to \cdots$ likewise runs off to infinity, and $J(e^z)$ is also the whole plane: in every neighbourhood of every point, however small, there are starting values with completely different behaviour.

The calculation also explains the power tower a, a^a, $a^{(a^a)}, \ldots$ for $a > 1$: with $\lambda = \ln a$ and $w = \lambda t$, the step $t \mapsto a^t$ becomes the mapping $w \mapsto \lambda e^w$. The tower therefore converges exactly for $\ln a \le 1/e$, that is, for $a \le e^{1/e} \approx 1.4447$.

> **Historical context** Pierre Fatou began iterating entire transcendental functions in 1926 and asked whether the Julia set of e^z can be the whole plane. Michał Misiurewicz proved this in 1981 [49]. By today's theory, every kind of Fatou component possible for e^z would have to "hold on to" the orbit of the special value 0; but this orbit escapes. The images of the Cantor bouquet and of the explosion come from work by Robert Devaney and co-authors from 1984 onwards.

8.5 Sine and cosine

Sine and cosine have critical points, at $\frac{\pi}{2} + k\pi$ for $\sin z$ and at $k\pi$ for $\cos z$, but only the critical values ± 1. So once again there are only a few "special" orbits.

- $f(z) = \sin z$: the fixed point 0 has $\sin'(0) = \cos 0 = 1$. It attracts real starting values only very slowly, much like the fixed point $\frac{1}{2}$ for the cauliflower $c = 0.25$ from Chapter 5. The Fatou set of $\sin z$ consists of points that tend to 0 in this way [42].
- $f(z) = \cos z$: pressing the cosine key on a calculator (in radians) again and again gives $x^* \approx 0.739085$, the only real fixed point of $\cos x = x$. Since $|\cos'(x^*)| = \sin x^* \approx 0.674 < 1$, it is attracting. It attracts both critical values ± 1, and the Fatou set is its basin of attraction.

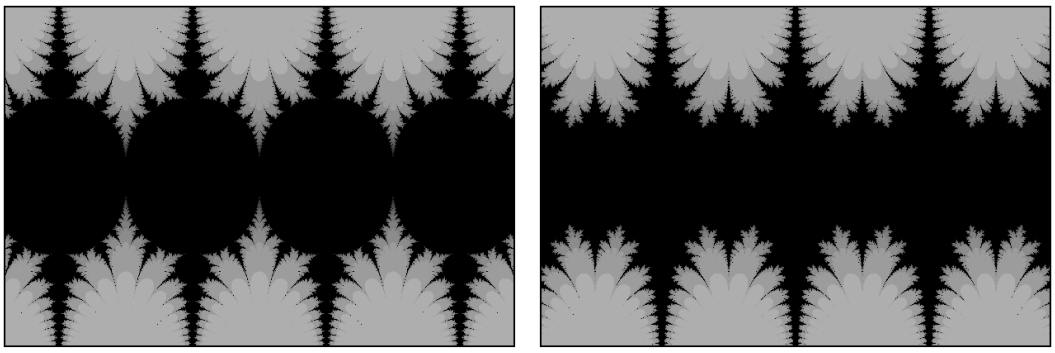


Figure 10: Escape-time images for $f(z) = \sin z$ (left) and $f(z) = \cos z$ (right). Light: $|\mathrm{Im}\, z|$ becomes large and the orbit escapes (the lighter, the faster). Black: the orbit remains bounded within the iteration limit. As for λe^z, the escaping, light points belong to the Julia set. The patterns repeat horizontally with period 2π.

Figure 10 again shows hairs, this time pointing up and down, because sine and cosine can

become large only in the vertical direction. The light regions, however, look not like thin curves but like areas.

This is not merely an effect of the image resolution: by McMullen, the Julia set of $\sin(\alpha z + \beta)$ has *positive area* for all $\alpha \neq 0$ and all β [33]. Since $\cos z = \sin(z + \frac{\pi}{2})$, this also holds for $\cos z$, even though the dynamics there, with an attracting fixed point, are as "calm" as can be. For $z^2 + c$ with an attracting fixed point, by contrast, the Julia set has area zero.

8.6 The tangent: a "fractal" that is a straight line

The tangent $\tan z = \sin z / \cos z$ has poles at $\frac{\pi}{2} + k\pi$; it is not entire but *meromorphic*. It has no critical points; its special values are $\pm i$: as $\operatorname{Im} z \to \pm\infty$, $\tan z$ tends to $\pm i$ without ever reaching these values.

We consider $\lambda \tan z$ with real $\lambda > 0$. Using the formulae from Section 8.3 one checks that $\operatorname{Im} \tan z$ has the same sign as $\operatorname{Im} z$. The upper and the lower half-plane are therefore each mapped into themselves. There the dynamics are orderly; chaos is possible only on the real axis. The behaviour there is described by Devaney and Keen, and by Keen and Kotus [50, 51]:

- For $0 < \lambda < 1$, 0 is an attracting fixed point ($(\lambda \tan)'(0) = \lambda < 1$). The Julia set lies on the real axis and is locally a Cantor set there. The Fatou set is a single connected region.
- For $\lambda > 1$ the Julia set is the whole real axis. There are two attracting fixed points $\pm i\xi_0$ with $\xi_0 = \lambda \tanh \xi_0$, for $\lambda = 2$ roughly $\xi_0 \approx 1.915$; the upper and lower half-planes are their basins of attraction.

In the limiting case $\lambda = 1$, 0 is a parabolic fixed point; here too the literature gives $J = \mathbb{R}$.

For $\lambda > 1$ the Julia set is therefore a straight line, the smoothest of all shapes, with Hausdorff dimension 1. The Julia set, however, is defined by the *dynamics*, not by its geometry: on it, small changes in the starting value lead to completely different orbits; near the poles the tangent flings neighbouring points arbitrarily far apart. Whether the set looks geometrically rough is a different question.

A similar example is the line segment $J_{-2} = [-2, 2]$ from Chapter 6. The case $0 < \lambda < 1$, on the other hand, corresponds to the case $c < -2$: a Cantor set on the real axis.

8.7 The logarithm: an orbit as a spiral

The logarithm is meant to solve $e^w = z$ for w. With $z = r\, e^{i\varphi}$ and $w = u + iv$ this means $e^u = r$ and $e^{iv} = e^{i\varphi}$, so

$$w = \ln r + i(\varphi + 2\pi k), \qquad k \in \mathbb{Z}.$$

Because of the period $2\pi i$ of e^z, every $z \neq 0$ has infinitely many logarithms; the complex logarithm is *multivalued*. For iterating, one chooses the *principal branch* at every step:

$$\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z \leq \pi.$$

It is undefined at $z = 0$ and jumps along the negative real axis; the theory of this chapter therefore cannot be applied directly. An orbit can even break off: since $\operatorname{Log} 1 = 0$, for $z_0 = 1$ the second step is already undefined.

Nevertheless, the iteration behaves regularly. Figure 11 shows the orbit of $z_0 = 2 + 0.5i$:

$$z_1 \approx 0.723{+}0.245i, \quad z_2 \approx -0.269{+}0.327i, \quad z_3 \approx -0.860{+}2.261i, \quad z_4 \approx 0.883{+}1.934i, \quad \ldots$$

It winds in a spiral towards the point

$$p \approx 0.3181 + 1.3372i,$$

which satisfies $\operatorname{Log} p = p$, that is, $e^p = p$.

The orbit cannot get away: the principal branch maps the upper half-plane into itself, because $0 < \operatorname{Arg} z < \pi$ holds there. By a classical theorem of complex analysis (the Denjoy–Wolff theorem), every orbit in the upper half-plane then converges to the only fixed point there. Orbits in the lower half-plane correspondingly tend to $\overline{p}$.

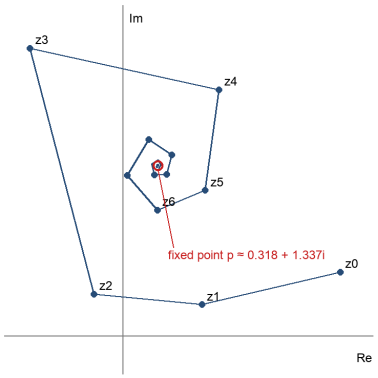


Figure 11: The orbit of the principal branch of the logarithm starting from $z_0 = 2 + 0.5i$. After a few steps it spirals into the fixed point $p \approx 0.318 + 1.337i$.

The spiral shape follows from the derivative. Close to p, Log acts like multiplication by $\operatorname{Log}'(p) = 1/p$: $z_{n+1} - p \approx \frac{1}{p}(z_n - p)$. By Section 1.6.2, the distance to p shrinks at each step by the factor $1/|p| \approx 0.7275$, and the connecting vector rotates by $-\arg p \approx -76.6°$. The result is a clockwise spiral with about 4.7 steps per revolution.

For e^z, p is also a fixed point, but there it is repelling, since $|(e^z)'(p)| = |e^p| = |p| \approx 1.375 > 1$. This fits with $J(e^z) = \mathbb{C}$, because repelling fixed points lie in the Julia set. Iterating the logarithm means iterating e^z backwards, and backwards, repelling points become attracting. Drawing Julia sets by backward iteration rests on the same principle.

8.8 Outlook: Apfelmännchen in unexpected places

The Mandelbrot set occurs far beyond $z^2 + c$. By McMullen, in every family of *rational* functions depending holomorphically on one complex parameter, the set of parameters at which the behaviour changes qualitatively is either empty or contains (distorted) copies of the boundary of a Multibrot set $\mathcal{M}_d$. Small Mandelbrot sets even lie densely there [52]. This is why, for example, Apfelmännchen are found in the parameter plane of Newton's method for cubic polynomials [19].

The theorem does not hold for the transcendental families of this chapter; their parameter planes are studied in their own right. For λe^z, the main cardioid corresponds to the set of λ with an

attracting fixed point, whose boundary passes through the explosion point $\lambda = 1/e$. In addition there are infinitely many regions with attracting cycles and, between them, "parameter hairs" [42]. The set at which the behaviour changes is, like $\mathcal{M}$, connected [53].

Exercises

1. **Euler's formula.** In Python, `1j` stands for i. Compare `cmath.exp(1j*phi)` with `math.cos(phi) + 1j*math.sin(phi)` for a few angles. The following program checks the period:

```
import cmath, math
z = 0.7 - 1.2j
print(cmath.exp(z), cmath.exp(z + 2j*math.pi))   # equal!
print(cmath.exp(z + 1j*math.pi))          # not equal: sign
```

   Explain why πi is not a period, even though $e^{z+\pi i}$ has the same modulus as e^z.

2. **The logarithm spiral.** Iterate the principal branch `cmath.log` starting from $2 + 0.5i$ and observe the distance to the fixed point p. The following program checks this:

```
import cmath
p = 0.3181315052 + 1.3372357014j
z = 2 + 0.5j
for n in range(1, 21):
    z = cmath.log(z)
    print(n, z, abs(z - p))
print(cmath.exp(p) - p)                   # p is a fixed point of exp
```

   Form the quotient of successive distances, determine its limit and compare it with $1/|p|$. Also examine the orbit with the starting value $2 - 0.5i$.

3. **Symmetry by hand.** For $d = 3$, compute the orbit terms 0, c, $c^3 + c$ for $c = 0.3 + 0.2i$ and for $c = -0.3 - 0.2i$, and explain why the second orbit is the reflection of the first through the origin (Section 8.1). Also determine, for $d = 4$, the parameter that gives an orbit rotated by $120°$.

9 Open questions

In Chapter 3 the Mandelbrot set $\mathcal{M}$ was introduced as the set of parameters c for which the orbit of $z_0 = 0$ under $f_c(z) = z^2 + c$ remains bounded. There followed the Julia sets J_c (Chapter 5), the alternative "connected or Cantor dust" for K_c (Chapter 6), the Hausdorff dimension (Chapter 7) and $z^d + c$, λe^z and other functions (Chapter 8). This chapter shows where established knowledge about these objects ends today.

The 1990 paper already touches such a limit. The "Historical context" box in Chapter 4 names a statement that was unproved at the time. Only some years later was it shown that the boundary of the Mandelbrot set has Hausdorff dimension 2 [31].

Each question is first described in simple terms and then stated precisely. Finally comes the *status as of October 2026*, with a source. A result counts as established only once its proof has been checked by experts and published in a journal (refereeing, known as *peer review*). Papers that so far exist only as a *preprint* on the arXiv server are explicitly identified as such.

9.1 Open questions about the Mandelbrot set and Julia sets

Is the Mandelbrot set locally connected? (MLC)

In simple terms. In 1982 Douady and Hubbard proved that $\mathcal{M}$ is connected [16]. The set consists of one piece, even though the small "Apfelmännchen" at the edge look like islands. Being *locally* connected is stronger: from every point one can move about within a small neighbourhood inside $\mathcal{M}$ without a large detour. MLC stands for "Mandelbrot set is locally connected".

Precise formulation.

For every $c \in \mathcal{M}$ and every neighbourhood U of c there is a neighbourhood $V \subseteq U$ of c such that $V \cap \mathcal{M}$ is connected.

Why it matters. Douady, Hubbard and Thurston built a combinatorial model of $\mathcal{M}$: a disc in which certain points are "pinched together" according to fixed rules. MLC states precisely that this model reproduces the actual Mandelbrot set and thus describes $\mathcal{M}$ completely [10]. Moreover, MLC implies the density of hyperbolicity (next section).

Status 2026: open. Around 1990 Yoccoz showed that $\mathcal{M}$ is locally connected at all parameters that are not "infinitely renormalisable". Roughly speaking, these points do not lie in infinitely many nested small copies of $\mathcal{M}$ [54]. For many remaining parameters there are partial results, most recently by Dudko and Lyubich [55]. A survey article for the 2026 International Congress of Mathematicians still treats MLC as an open problem [56] (preprint).

Density of hyperbolicity

In simple terms. The interior of $\mathcal{M}$ contains the main cardioid and countless discs and bulbs (Section 3.5). For c in these regions the orbit of 0 is attracted by a cycle, such as a fixed point or a 2-cycle. Such parameters are called *hyperbolic*. It is open whether they are dense everywhere in $\mathcal{M}$ or whether the interior also contains regions of a different kind.

Precise formulation.

The set of $c \in \mathbb{C}$ for which f_c has an attracting cycle is dense in $\mathcal{M}$.

Equivalently: every component of the interior of $\mathcal{M}$ is hyperbolic.

Status 2026: real case solved, complex case open. For real $c \in [-2, \frac{1}{4}]$, that is, for $x \mapsto x^2 + c$ on the number line, Lyubich and, independently, Graczyk and Świątek proved density in 1997 [57, 58]. Kozlovski, Shen and van Strien extended this in 2007 to all real polynomials [59]. For complex c the question is open; MLC would answer it.

How large is the area of $\mathcal{M}$?

In simple terms. $\mathcal{M}$ lies in a rectangle of roughly 2.5×2.5 units. No formula for its exact area is known.

What is known. Counting, on a very fine grid, the points whose orbit does not escape gives

$$\text{area}(\mathcal{M}) \approx 1.50659.$$

This is a *numerical estimate* from very large computations, not refereed and not a proof [60]. Only bounds have been proved. In 1993 Fisher and Hill stated in an unpublished manuscript [61]:

$$1.50296686 < \text{area}(\mathcal{M}) < 1.57012937.$$

A series expansion yields an exact formula for the area. Its truncations give upper bounds that approach the true value only very slowly [62].

Status 2026: open. An exact value is not known.

Does the boundary of $\mathcal{M}$ have area zero?

In simple terms. The boundary of a circle is a line without area. The boundary of $\mathcal{M}$, by contrast, is so jagged that it has Hausdorff dimension 2 [31], and is thus in this sense "as large as a surface" (Chapter 7). A set of dimension 2 can nevertheless have area zero.

Precise formulation.

$$\text{Is} \quad \text{area}(\partial\mathcal{M}) = 0 \ ?$$

Status 2026: open [10]. The question also concerns pixel counting. If the boundary had positive area, there would be a "blurred" portion that no picture can cleanly assign to the interior or the exterior.

A solved question: Julia sets of positive area

In simple terms. All the Julia sets J_c in the gallery in Chapter 5 are thin structures without area, curves or dust. For a long time it was open whether this holds for *every* Julia set of $z^2 + c$.

Precise formulation.

Is there a $c \in \mathbb{C}$ for which J_c has positive area?

Status: solved, the answer is yes. In 2012 Buff and Chéritat constructed quadratic polynomials whose Julia set has positive area [63]. In 2022 Avila and Lyubich found further examples, the so-called Feigenbaum Julia sets [64].

9.2 Open questions in fractal geometry

The following questions do not concern $z^2 + c$, but they do concern the tools from Chapter 7: dimension, coverings and self-similarity.

The Kakeya problem

In simple terms. In 1917 Kakeya asked how little area one needs in order to turn a needle of length 1 in the plane once completely round. Besicovitch showed that there are sets that contain a line segment of length 1 in *every* direction and yet have area (in space: volume) 0. According to the conjecture, these Kakeya sets are nevertheless not "thin": their dimension is full.

Precise formulation.

Every compact set $K \subseteq \mathbb{R}^n$ that contains a line segment of length 1 in every direction has Hausdorff dimension n.

Status 2026: proved in the plane; in space a generally accepted proof (main article still a preprint); open from dimension 4 upwards. For $n = 2$ the statement has been proved since 1971 (Davies). For $n = 3$ Hong Wang and Joshua Zahl presented a proof in February 2025 [65]. The main article has so far appeared only as a preprint. Important building blocks, however, have been published after refereeing [66], a simplified version has been accepted for publication [67], and experts consider the proof to be correct [68]. On 23 July 2026 Hong Wang received the Fields Medal, one of the highest distinctions in mathematics, in part for this work [69]. For $n \geq 4$ the conjecture is open.

Falconer's distance problem

In simple terms. Consider all distances between pairs of points of a set E in the plane. If E is "large enough", these distances should fill a whole stretch of the number line. In 1985 Falconer conjectured that a dimension greater than 1 is already sufficient for this.

Precise formulation.

$$E \subseteq \mathbb{R}^2 \text{ compact, } \dim_{\mathrm{H}} E > 1 \implies \Delta(E) = \{\, |x - y| : x,y \in E \,\} \text{ has positive length.}$$

In $\mathbb{R}^n$ the threshold is $n/2$ instead of 1.

Status 2026: open. In the plane it has been proved that $\dim_{\mathrm{H}} E > \frac{5}{4}$ suffices (Guth, Iosevich, Ou and Wang, 2020) [70]. The gap between 1 and $\frac{5}{4}$ has not yet been closed.

The exact overlaps conjecture

In simple terms. The Cantor set consists of two reduced, non-overlapping copies of itself. Its dimension therefore follows from the formula in Section 4.2. If the copies overlap, the dimension

can be smaller. According to the conjecture, on the number line this happens *only if* two copies coincide exactly; chance near-overlaps do no harm.

Precise formulation. Let $S_i(x) = r_i x + t_i$ with $0 < |r_i| < 1$ be finitely many similarity maps with the self-similar set F and the similarity dimension s, that is, $\sum_i |r_i|^s = 1$. If no two distinct compositions $S_{i_1} \circ \cdots \circ S_{i_k}$ and $S_{j_1} \circ \cdots \circ S_{j_l}$ are equal, then

$$\dim_{\mathrm{H}} F = \min\{1, s\}.$$

Status 2026: on the number line a proof claimed, not yet checked. Hochman proved the statement in 2014 under an additional separation condition [71], and Rapaport in 2022 for algebraic contraction factors [72]. On 7 October 2026, eleven days before the date of this text, Kittle and Kogler published a proof of the full one-dimensional case as a preprint, together with a computer check in the proof assistant Lean [73]. According to their account, an AI system (GPT-6 Astra) found the first proof; an independent proof produced by OpenAI had become public one day earlier. The paper is *not refereed*. A proof is therefore claimed, but it is not yet established. In higher dimensions the question is open.

Furstenberg's $\times 2 \times 3$ conjecture

In simple terms. Multiplying a number from $[0, 1)$ by 2 and discarding the integer part shifts its binary representation by one place. Multiplying by 3 likewise shifts the representation in base 3. Furstenberg conjectured that the two operations are so "independent" that nothing interesting fits both at once.

Precise formulation.

Every ergodic probability measure on $[0, 1)$ *that is invariant under* $x \mapsto 2x \bmod 1$ *and under* $x \mapsto 3x \bmod 1$ *is either the ordinary length measure or is supported on finitely many rational numbers.*

Status 2026: open; a related conjecture is solved. In 1990 Rudolph proved the case of "positive entropy" [74]; the rest is open. A related conjecture of Furstenberg on the dimension of intersections of $\times 2$- and $\times 3$-invariant sets was proved in 2019 by Shmerkin and, independently, by Wu [75, 76].

9.3 How conjectures end

A conjecture ends with a proof or with a counterexample. Fractal geometry has an example of each.

Proved: the frontier of Brownian motion

In simple terms. A particle that jumps randomly back and forth in the plane traces out a tangled path (Brownian motion). Its outer boundary is a jagged curve. On the basis of computer pictures, Mandelbrot conjectured that this boundary has dimension $\frac{4}{3}$.

Question	since	Status October 2026
Local connectivity of $\mathcal{M}$ (MLC)	1980s	open
Density of hyperbolicity	1920s	real case solved (1997), complex open
Area of $\mathcal{M}$	c. 1990	open, only bounds
Area of $\partial\mathcal{M}$ equal to 0?	1980s	open
Julia sets of positive area	1980s	solved: yes (2012)
Kakeya conjecture	1917	$n = 2$ solved, $n = 3$ proved (preprint), $n \geq 4$ open
Falconer's distance problem	1985	open ($\frac{5}{4}$ reached)
Exact overlaps	2014	$\mathbb{R}$: proof claimed (preprint)
Furstenberg's $\times 2$ $\times 3$ conjecture	1967	open

Table 2: The open questions of this chapter at a glance. The column "since" indicates only approximately since when the question has been posed in its present form.

Precise formulation. With probability 1, for the path $B[0,1]$ of a planar Brownian motion,

$$\dim_{\mathrm{H}}\big(\text{outer boundary of } B[0,1]\big) = \tfrac{4}{3}.$$

Result. Lawler, Schramm and Werner proved this in 2001 with a method that was new at the time, the Schramm–Loewner evolution [36]. This work was among the reasons for Werner's Fields Medal in 2006. Here a conjecture drawn from pictures turned out to be correct.

Refuted: Eremenko's conjecture

In simple terms. For functions such as λe^z from Chapter 8, the *escaping set* $I(f)$ consists of the points whose orbit runs off to infinity. In the pictures it looks like hairs that reach all the way to infinity. In 1989 Eremenko asked whether this holds for every connected piece of $I(f)$ [77].

Precise formulation.

For every transcendental entire function f, every connected component of $I(f) = \{z : f^n(z) \to \infty\}$ is unbounded.

Result. A stronger version (every escaping point can be joined to ∞ by a curve in $I(f)$) was refuted in 2011 by Rottenfußer, Rückert, Rempe and Schleicher [78]. The conjecture itself was refuted by Martí-Pete, Rempe and Waterman; their paper appeared in 2025 in the *Journal of the American Mathematical Society* [79]. They construct an entire function whose escaping set has a bounded piece. According to an explanatory note (preprint, September 2026), such a piece can even be a single point [80]. Part of the original idea remains true: Rippon and Stallard had shown in 2005 that $I(f)$ always has *at least one* unbounded piece [81].

For the exponential functions λe^z even the stronger version holds [47]; there the hairs in the pictures do not deceive. The counterexamples are specially constructed, very particular functions. Pictures alone therefore prove nothing: a conjecture that holds for all known examples can be false.

9.4 Further exploration

Like the pictures in the 1990 paper, many questions in this chapter arose from experiments on the computer. Today a short program produces in seconds pictures that required long computing times in 1990, and numerical values can be recomputed independently.

New results usually appear first, freely accessible, on https://arxiv.org; for complex dynamics, for instance, in the category *Dynamical Systems* (math.DS). Popular-science magazines carry readily understandable reports on new proofs, for example *Quanta Magazine* (https://www.quantamagazine.org).

The distinction made in this chapter remains important here: a preprint is a *proposed* proof. Only careful checking by others turns it into established knowledge, and that sometimes takes years.

Exercises

1. **The area of $\mathcal{M}$ by pixel counting.** The following program lays a grid over the upper half of $\mathcal{M}$ and counts the grid points whose orbit has not escaped after 200 steps. Because of the symmetry about the real axis, the result is doubled.

```
N, maxit = 400, 200                      # grid points per axis, iterations
x0, x1, y0, y1 = -2.0, 0.5, 0.0, 1.25    # upper half suffices (symmetry)
dx, dy = (x1 - x0) / N, (y1 - y0) / N
hits = 0
for i in range(N):
    for j in range(N):
        c = complex(x0 + (i + 0.5) * dx, y0 + (j + 0.5) * dy)
        z = 0
        for n in range(maxit):
            z = z * z + c
            if abs(z) > 2:
                break
        else:
            hits += 1
print("Area approximately", 2 * hits * dx * dy)
```

   With these values one obtains about 1.527. Compare this with the estimate 1.50659 and the proved bounds from Section 9.1, and vary `N` and `maxit`. Explain why the result tends to be too high, by considering the points that the program counts even though they do not belong to $\mathcal{M}$.

2. **Reading a report.** Find an article in *Quanta Magazine* on the Kakeya problem or the Mandelbrot set, note which statements are presented there as proved and which as conjectured, and compare with Table 2.

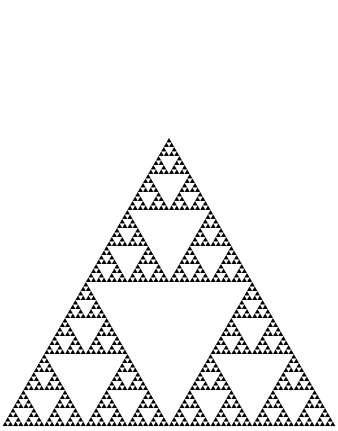
3. **Today's research.** On arXiv, open the list of new papers in the category math.DS (*Dynamical Systems*), choose a title containing "Mandelbrot" or "Julia" and read the summary (*abstract*). Check whether the paper has already appeared in a journal; if so, this is stated in the *Journal reference* field.

10 Appendix

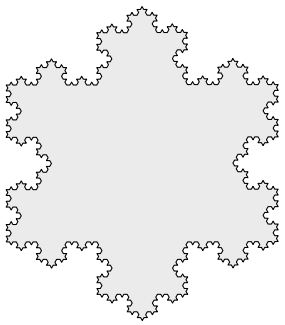
10.1 Various images

The following plates preserve the original selection of images and, where the original gives them, the coordinates and iteration depths. The Mandelbrot images with coordinates were recomputed in black and white with a Python program. The three-dimensional representation was likewise regenerated from the escape times. The colour plate was replaced by similar details, and the Koch curve and the scaling table were redrawn in LaTeX. The plate of IFS fractals shows three newly generated examples. The explanations beneath them make the mathematical meaning more precise.

Plate 1: Examples of iterated function systems (original page 31)



Sierpiński triangle



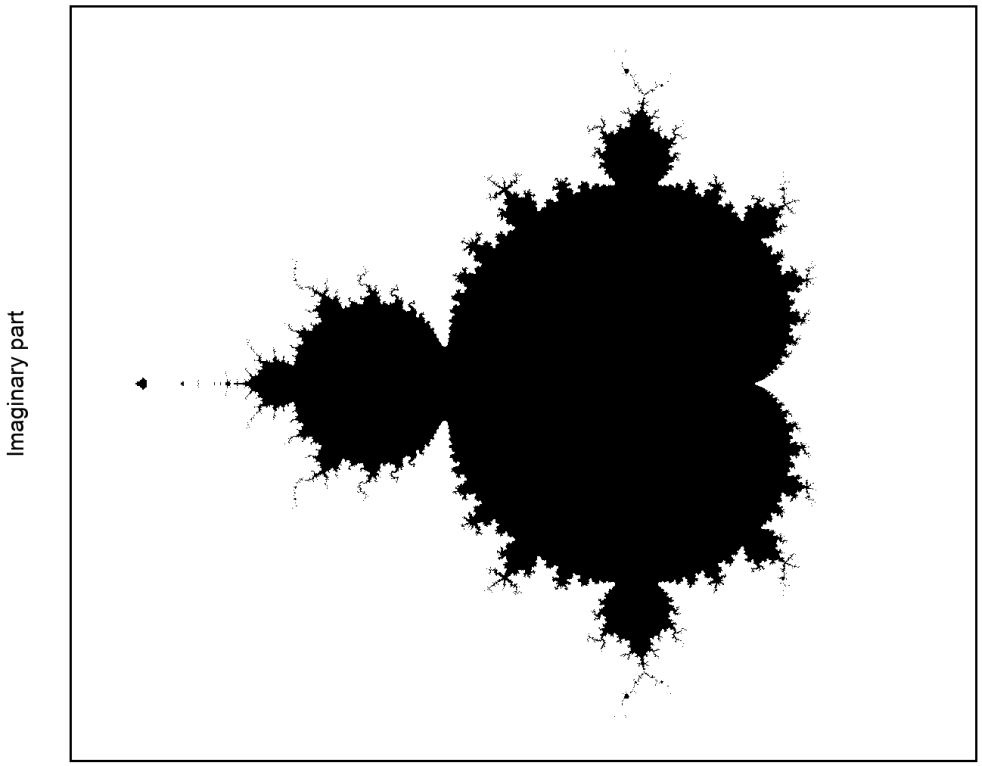
Koch snowflake



Barnsley fern

The original shows various IFS fractals, among them the Koch snowflake and the Sierpiński triangle. The remaining motifs lack labels or source references. The images were regenerated in 2026 and are restricted to examples with a known rule: the Sierpiński triangle and the snowflake are constructed exactly, and the fern is produced by the four affine maps given by Michael Barnsley (*Fractals Everywhere*, 1988). These figures are produced by iterated function systems. They do not belong to the Mandelbrot set, but they likewise show how complicated, self-similar structures can arise from a simple, repeated rule.

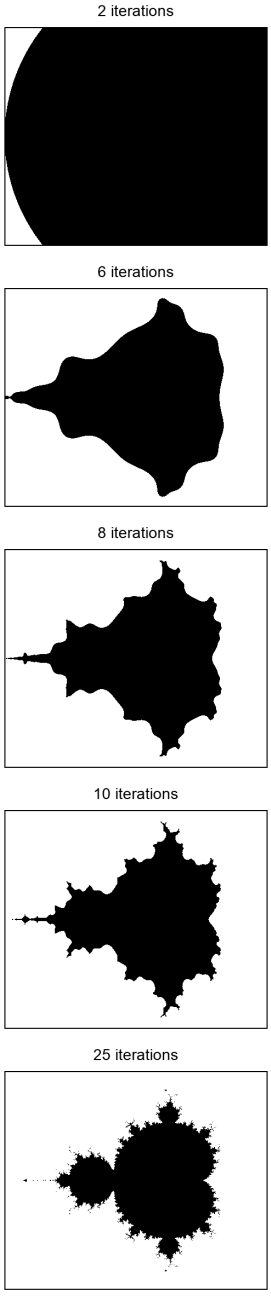
Plate 2: The Mandelbrot set in the complex plane (original page 32)




```
   Real part:  Minimum -2.00000000   Maximum 1.00000000
Imaginary part:  Minimum -1.25000000   Maximum 1.25000000
                   Iteration depth: 25
```

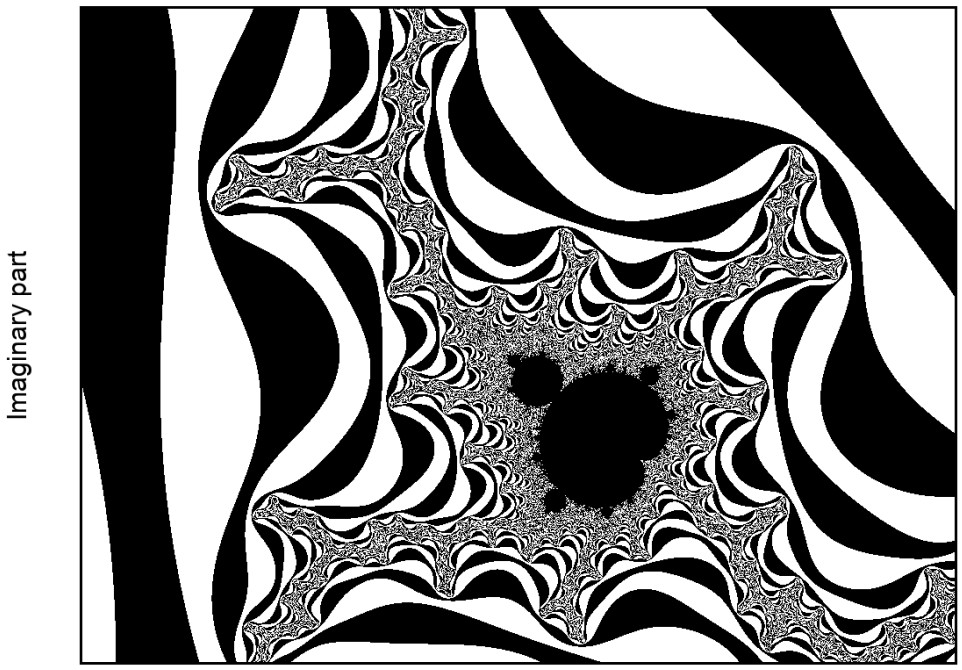
The black points have not escaped after 25 iterations. The picture is therefore a numerical approximation of the Mandelbrot set. A black point need not ultimately belong to the Mandelbrot set; its sequence may cross the escape boundary only after more than 25 steps. Convergence to zero is not assumed.

Plate 3: Effect of the iteration limit (original page 33)



The sequence of images shows the approximation after 2, 6, 8, 10 and 25 iterations. As the iteration limit rises, more and more points lying outside are recognised as escaping. The Mandelbrot set itself does not change; only the outer approximation becomes more accurate. Ever finer details therefore become visible, especially at the boundary, as the iteration depth increases.

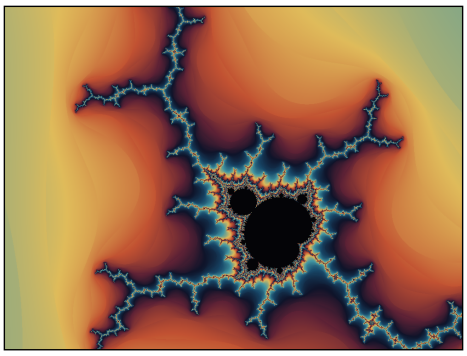
Plate 4: Magnification of a boundary region (original page 34)



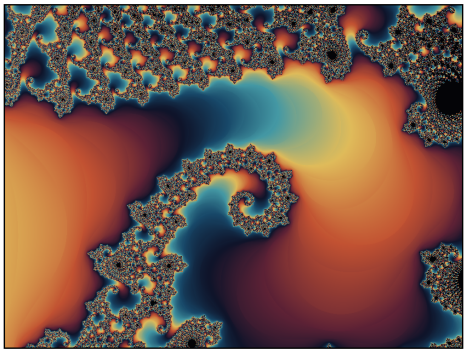

```
  Real part:  Minimum -0.19920000   Maximum -0.12954000
Imaginary part:  Minimum 1.01480000   Maximum 1.06707000
                 Iteration depth: 200
```

The image shows a magnification at the boundary of the Mandelbrot set. Further magnification brings out ever new structures and miniature copies. The region shown is only 0.06966 wide in the real part. The black and white bands denote different escape times; they are not additional parts of the Mandelbrot set.

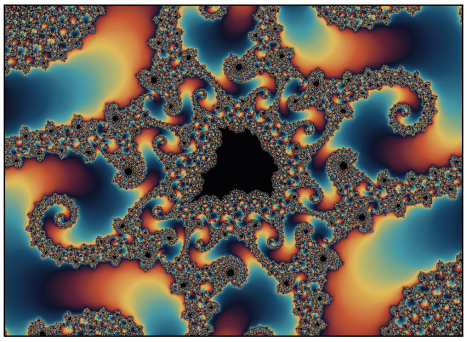
Plate 5: Coloured fractal images (original page 35)



Real part: Minimum −0.19920000 Maximum −0.12954000
Imaginary part: Minimum 1.01480000 Maximum 1.06707000
Iteration depth: 260



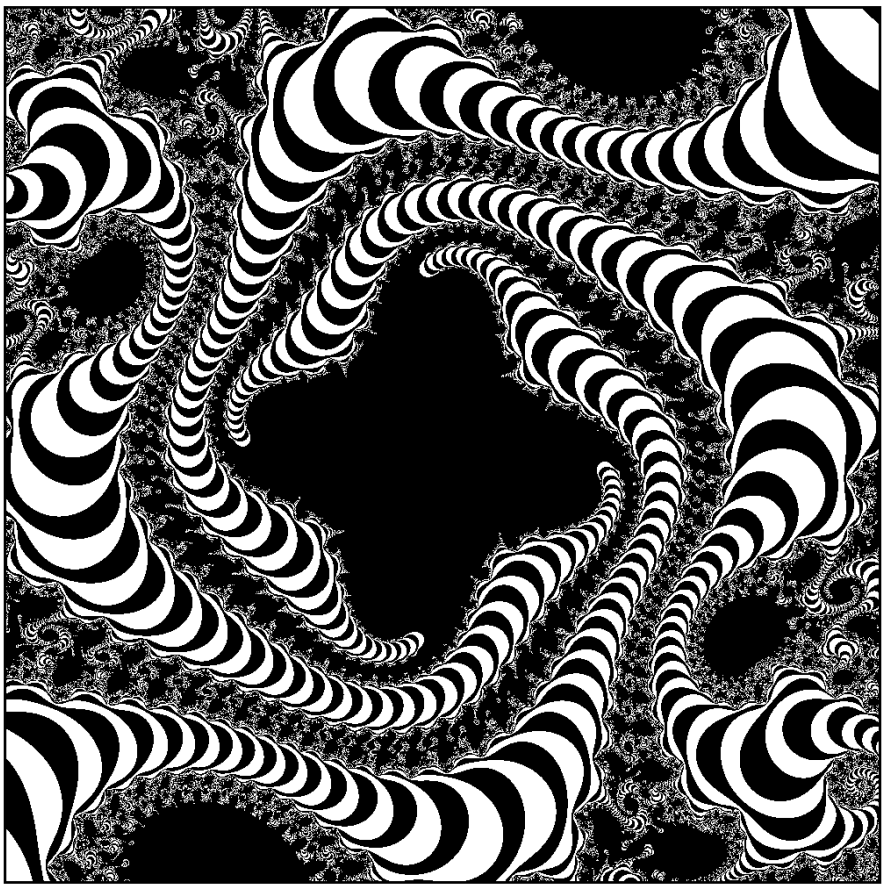
Real part: Minimum −0.74700000 Maximum −0.74000000
Imaginary part: Minimum 0.12900000 Maximum 0.13425000
Iteration depth: 320



Real part: Minimum −0.74543560 Maximum −0.74542150
Imaginary part: Minimum 0.11300370 Maximum 0.11301390
Iteration depth: 450

The original colour sheet survives in scan `IMG_1960` but gives no coordinates. For this reading edition, three similar details were therefore recomputed with a Python program; no claim is made to an exact reconstruction of the original images. The coordinate range used and the iteration depth are given beneath each image. The colours represent groups of similar escape times. They are not part of the Mandelbrot set as a mathematical set of points. A different colour assignment would therefore change the appearance, but not the computed sequences.

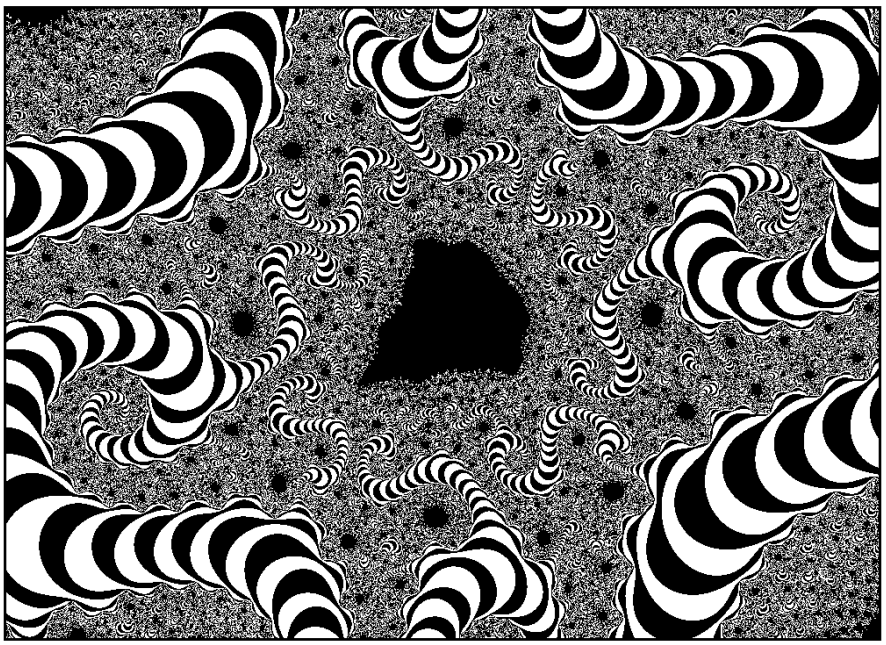
Plate 6: Black-and-white recomputation (original page 36)




```
   Real part:  Minimum -0.74638270   Maximum -0.74638210
Imaginary part:  Minimum -0.09850600   Maximum -0.09850540
                 Iteration depth: 350
```

A strong magnification with 350 iterations. The narrow black connections and recurring miniature copies illustrate the fine structure of the boundary. The width of the detail in the real part is only 0.00000060. The repetitions are similar to the main figure but not identical to it, since new structures appear at every level of magnification.

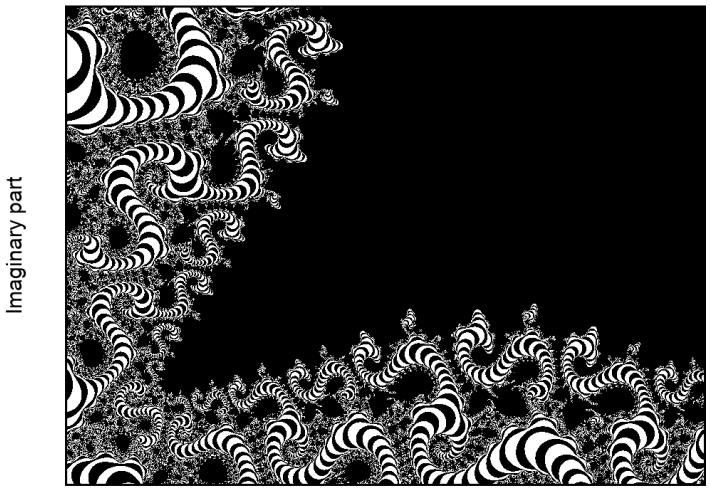
Plate 7: Another miniature copy (original page 37)



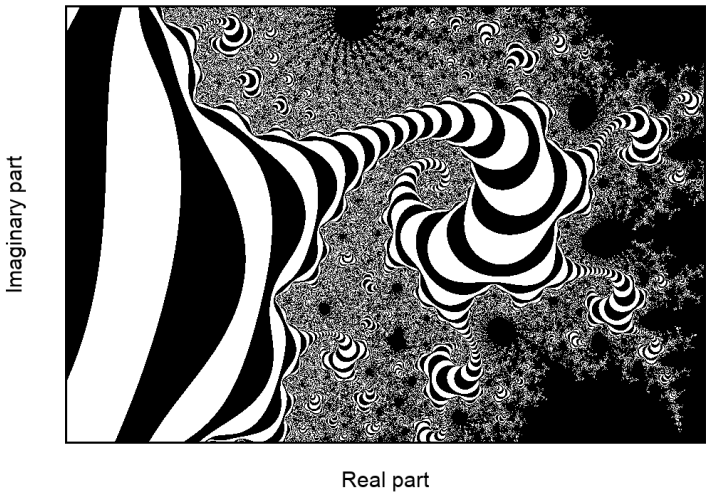

```
  Real part:  Minimum -0.74543560   Maximum -0.74542150
Imaginary part:  Minimum 0.11300370   Maximum 0.11301390
                 Iteration depth: 350
```

This detail, too, shows an approximate symmetry and a point-like miniature copy of the main figure. Its width in the real part is 0.00001410. The Mandelbrot set as a whole is exactly symmetric about the real axis. The local symmetry visible here, by contrast, is only approximate.

Plate 8: Two magnified details (original page 38)



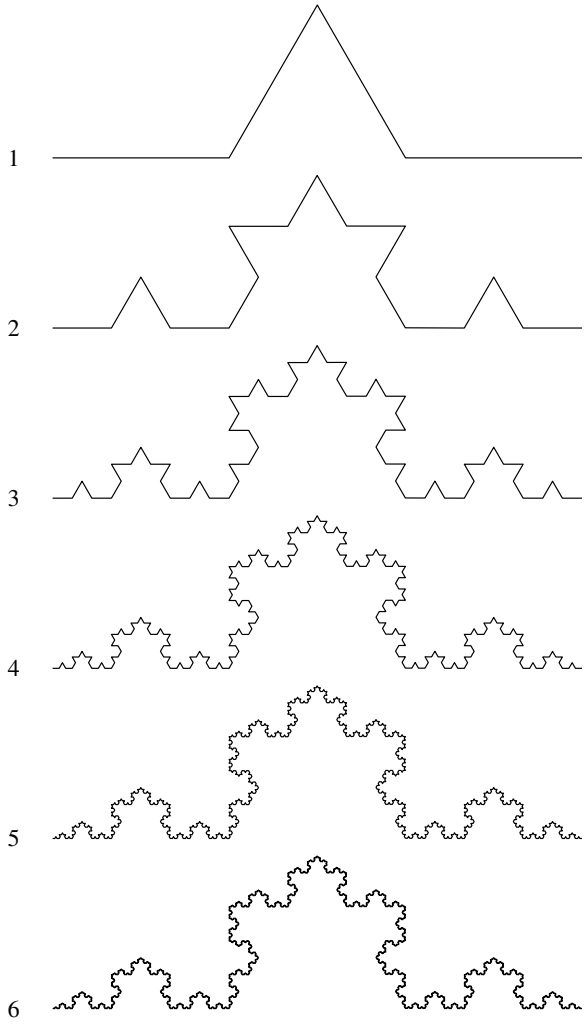

```
  Real part:  Minimum -0.74543010   Maximum -0.74542890
Imaginary part:  Minimum 0.11300760   Maximum 0.11300850
                 Iteration depth: 350
```

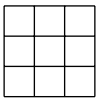
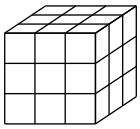



```
  Real part:  Minimum -0.75104000   Maximum -0.74080000
Imaginary part:  Minimum 0.10835000   Maximum 0.11536000
                 Iteration depth: 300
```

The images document different details and iteration depths of the computer representation of the time. The upper detail spans only 0.0000012 in the real part, the lower one 0.01024. The bands arise from showing the escape times alternately in black and white. They make the speed of escape visible, but they do not prove that an individual point belongs to the Mandelbrot set.

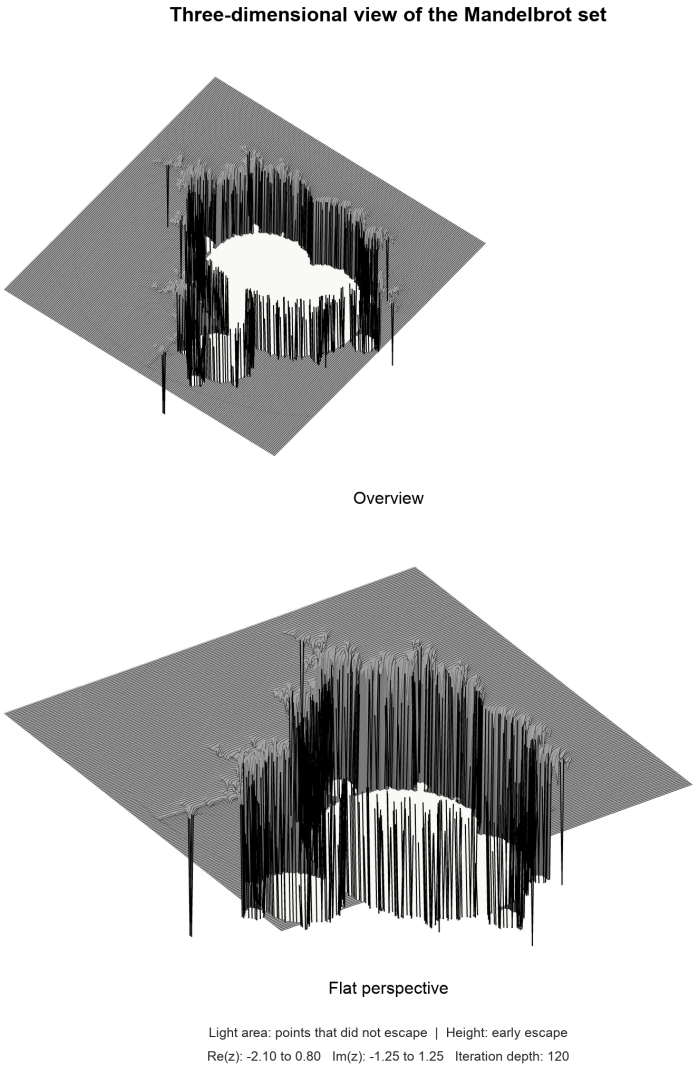
Plate 9: Koch curve and scaling (original page 39)



	Shape	Number N	Scaling s	Law
	Line segment	3	3	$3^1 = 3$
	Square	9	3	$3^2 = 9$
	Cube	27	3	$3^3 = 27$

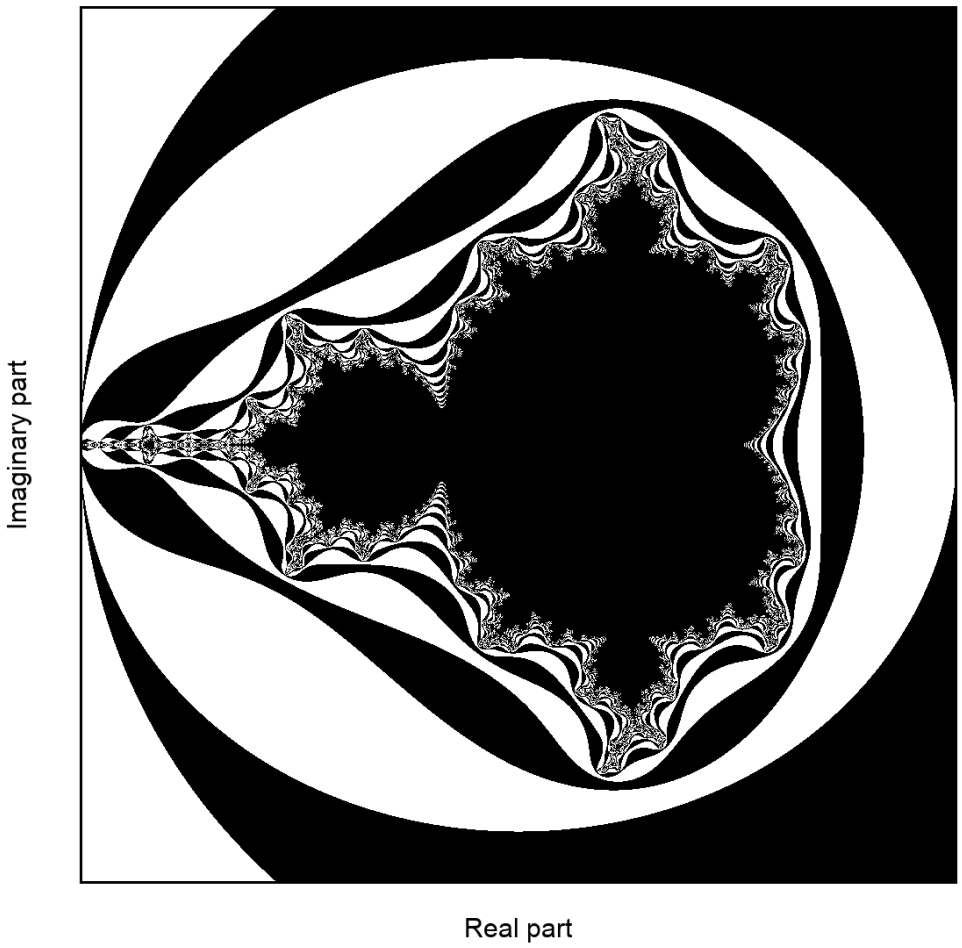
The iteration stages of the Koch curve and the table for line segment, square and cube illustrate the scaling law $N = s^D$. The curve and the table were redrawn from the original in 2026; the original's symbols a and d have been brought into line with the notation $N = s^D$ used in Section 4.2. The Koch curve is generated by a different rule from the Mandelbrot set. It serves here as a clear example on which self-similarity and a fractional dimension can be calculated directly.

Plate 10: Three-dimensional representation (original page 40)



The two perspectives were recomputed for $\mathrm{Re}(c) \in [-2.10, 0.80]$, $\mathrm{Im}(c) \in [-1.25, 1.25]$ and an iteration depth of 120; the original gives no coordinates for this plate, and the values were chosen in 2026. Points whose orbit does not escape within this limit form the light-coloured surface of the "Mandelbrot lake". For the remaining points, the height is chosen to be greater the earlier the orbit crosses the escape radius. The height is therefore not a third spatial coordinate of the Mandelbrot set but a visualisation of numerical data. Depending on the height function, the same computation can produce landscapes that look different; the planar Mandelbrot set remains unchanged.

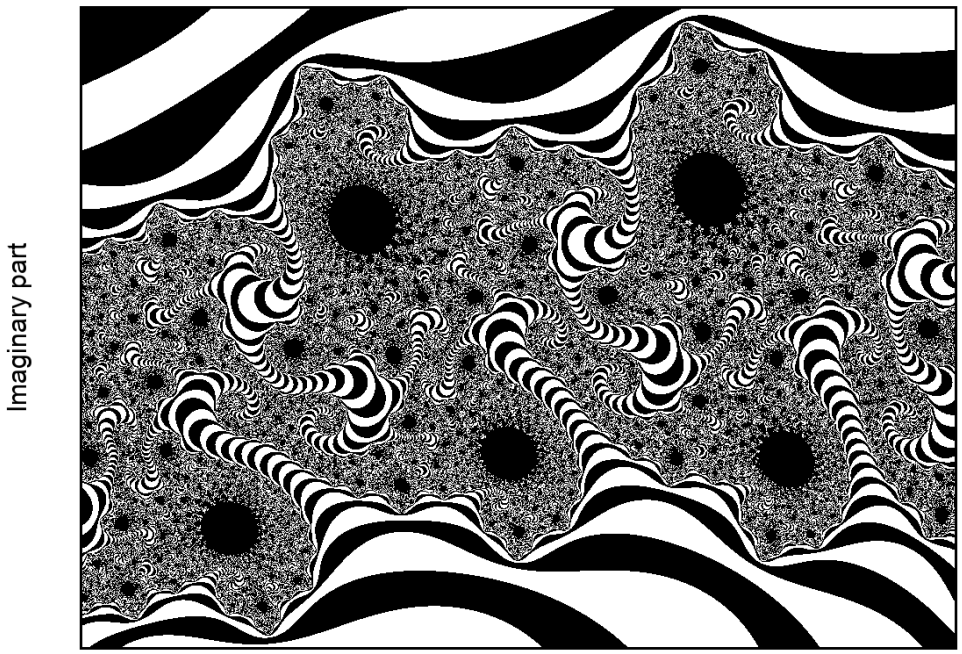
Plate 11: Differentiated graphical representation (original page 41)




```
   Real part:  Minimum -2.00000000   Maximum 1.00000000
Imaginary part:  Minimum -1.50000000   Maximum 1.50000000
                 Iteration depth: 25
```

The representation uses the differentiated colouring or shading explained in Section 3.4. Points with similar escape times are grouped into common bands. The outer rings and arcs thus show not only the boundary of the Mandelbrot set but also the behaviour of the escaping sequences in its neighbourhood.

Plate 12: Detail with high iteration depth (original page 42)




```
  Real part:  Minimum −0.74553800   Maximum −0.74505400
Imaginary part:  Minimum 0.11288100   Maximum 0.11323600
                   Iteration depth: 250
```

The detail was produced with an iteration depth of 250. Its width in the real part is only 0.000484. Despite this strong magnification, miniature copies and branching boundary structures appear once again. One can therefore speak of approximate self-similarity; an exact copy of the whole figure, however, does not occur.

10.2 Afterword

Are fractals merely mathematical games, or do they also prove their worth in practice? Fractal methods are in fact applied in various fields. Algorithms consisting of only a few instructions can generate artificial landscapes that are barely distinguishable from real ones. Such methods can be used in film production, for example, to supplement or replace elaborate models.

A further application lies in digital image transmission and image compression. Image information can be described compactly by suitable mathematical rules. This allows data to be transmitted faster, which matters particularly where storage space and transmission time are scarce or expensive.

The last chapters show that fractal geometry is an extensive field within mathematics. A Facharbeit can give only a general insight. The limits of school mathematical knowledge are quickly reached, since school mathematics is not sufficient for many of the details. The references list several works that deal with particular fractal structures and related topics.

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Note on sources The list follows the order and selection of the original scan. In 2026 the details were checked against library catalogues and publishers’ information and supplemented with subtitles and series. Entry [3] has been corrected: *The Science of Fractal Images* is a volume edited by Peitgen and Saupe and appeared in 1988, not 1986. The original gives no page numbers for the individual statements and figures used. Entries [1] to [9] form the 1990 list; from [10] onwards follow the sources used in 2026 for the further chapters 5 to 9.

10.4 Pascal listing

The printout of the program runs to eleven sheets: the first begins with `PROGRAM FRAKTAL;`, the last ends with `END.`. The following listing is a character-for-character transcription of this printout from 1990: comments, umlauts, spellings (including typing errors), brackets and indentation are reproduced as printed. Only lines that the printer had wrapped after 80 characters have been rejoined; the character hidden at the edge of the paper in the process was supplied from the context. The comments and on-screen texts in the listing are in German, as printed in 1990.

A complete compilation is nevertheless not possible, because the included files `GEMCONST.PAS`, `TRIXCONST.PAS`, `GEMTYPE.PAS`, `TRIXTYPE.PAS`, `GEMSUBS` and `TRIXSUBS` are not available. Routines such as `PAINT_RECT`, `GET_SCREEN` and `INIT_GEM` also come from them. The listing therefore documents the accessible source code, but without these contemporary libraries it is not a self-contained, compilable program.

Original Pascal source code from 1990

```
PROGRAM FRAKTAL;

 CONST

  {$I GEMCONST.PAS}         { Quelltext - Konstanten             }
  {$I TRIXCONST.PAS}        { Zusätzliche Quelltext - Konstanten }

 TYPE

  {$I GEMTYPE.PAS}          { Quelltext - Datentypen             }
  {$I TRIXTYPE.PAS}         { Zusätzliche Quelltext - Datentypen }

 VAR TASTE         : CHAR;           { einfacher Character                }
     I             : INTEGER;        { Wird im Zusammenhang mit Scan ver-
                                       wendet. Entspricht dem Zahlenwert
                                       der jeweiligen Taste                 }

     BILDSCHIRM    : SCREEN_TYPE;    { ST spezielles Bildschirm - Fomat    }
     BILD_SPEICHER : BOOLEAN;        { überprüft ob sich ein Bild im
                                       Speicher befindet. false entspricht
                                       einem leeren und true einem vollen
                                       Bildschirm-Speicher.                 }
     EN            : BOOLEAN;

  {$I GEMSUBS}              { Quelltext - Prozedur- und Funktionsköpfe }
  {$I TRIXSUBS}             { Weitere Prozedur- und Funktionsköpfe     }


 { Diese Routine fragt die Scancodes der Tastatur ab. Auf Grund dieser
   Routine ist es möglich auch spezielle Tasten (z.B. Funktionstasten)
   abzufragen. Dies ist mit dem herkömmlichen readln - Befehl nicht
   möglich.                                                          }

 FUNCTION BCONCIN(DEV : INTEGER) : LONG_INTEGER;BIOS(2);

 FUNCTION SCAN : INTEGER;

  BEGIN
   SCAN:=INT(BCONCIN (2) DIV (65535))
  END;

 { Prozedur zur Kursorsteuerung }

 PROCEDURE CURSOR(FUNKTION,RATE : INTEGER);XBIOS(21);
```

```
45
46
47  {-------------------------------------------------------------------------
48   FRACTALE AUSDRUCKEN
49   -------------------------------------------------------------------------}
50
51  PROCEDURE DRUCKEN;
52
53   CONST X_MAX = 639;
54         Y_MAX = 399;
55
56
57   VAR X,Y,I,J,CODE : INTEGER;
58       OK           : ARRAY[1..8] OF BOOLEAN;
59
60   BEGIN
61    IF (BILD_SPEICHER = FALSE) THEN
62     BEGIN
63      CUR_OFF;
64      CLEAR_SCREEN;
65      HIDE_MOUSE;
66      PAINT_STYLE(0);
67      PAINT_RECT(1,1,638,398);
68      PAINT_STYLE(1);
69      PAINT_RECT(4,4,632,392);
70      PAINT_STYLE(5);
71      PAINT_RECT(5,5,630,390);
72      PAINT_STYLE(1);
73      PAINT_RECT(232,189,194,34);
74      PAINT_STYLE(0);
75      PAINT_RECT(228,185,190,30);
76      GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
77      I:=SCAN;
78     END
79    ELSE
80     BEGIN
81      PUT_SCREEN(BILDSCHIRM);
82      REWRITE(OUTPUT,'PRN:');    { Initialisierung des Druckers }
83      WRITELN(CHR(27),'U1');
84      WRITELN(CHR(27),'3',CHR(16));
85      X:=0;
86      Y:=0;
87      REPEAT
88       write(chr(27),'*',chr(6),chr((x_max+40+1) mod(256)),chr((x_max+40+1)div(256)));
89       for j:=1 to 40 do write(chr(0)); {linker Rand}
90       for x:=0 to x_max do
91        BEGIN
92         FOR I:=1 TO 8 DO OK[I]:=FALSE;
93         FOR I:=1 TO 8 DO
94          BEGIN
95           IF GET_PIXEL(X,Y+I-1)=BLACK THEN OK[I]:=TRUE;
96           IF I=8 THEN
97            BEGIN
98             CODE:=0;
99             IF OK[1]=TRUE THEN CODE:=CODE+128;
100            IF OK[2]=TRUE THEN CODE:=CODE+64;
101            IF OK[3]=TRUE THEN CODE:=CODE+32;
102            IF OK[4]=TRUE THEN CODE:=CODE+16;
103            IF OK[5]=TRUE THEN CODE:=CODE+8;
104            IF OK[6]=TRUE THEN CODE:=CODE+4;
105            IF OK[7]=TRUE THEN CODE:=CODE+2;
106            IF OK[8]=TRUE THEN CODE:=CODE+1;
107            WRITE(CHR(CODE));
108            IF X = X_MAX THEN FOR J:=1 TO (90*8-(X_MAX+1)-40) DO WRITE(CHR(0));
109           END;
110         END;
111       END;
```

```
      WRITELN(CHR(27),'J',CHR(8));
      Y:=Y+8;
    UNTIL Y>Y_MAX+8;
    REWRITE(OUTPUT,'CON:');
   END;
 END;

{-------------------------------------------------------------------------
 FRACTALE BERECHNEN
 -------------------------------------------------------------------------}

PROCEDURE BERECHNEN;

 VAR X_KOORDINATE,                  { x - Koordinate }
     Y_KOORDINATE,                  { y - Koordinate }
     RECHENTIEFE,                   { Rechentiefe    }
     ZAEHLER,                       { Zähler für Iterationen }
     X_Z,                           { Zähler für x - Koordinate }
     Y_Z           : INTEGER;       { Zähler für y - Koordinate }
     REAL_MIN,                      { Realteil - Minimum }
     REAL_MAX,                      { Realteil - Maximum }
     IMAG_MIN,                      { Imaginärteil - Minimum }
     IMAG_MAX,                      { Imaginärteil - Maximum }
     X_SCHRITT,                     { Schrittweite in x - Richtung }
     Y_SCHRITT,                     { schrittweite in y - Richtung }
     I_KONST,                       { Imaginärteil der Konstante }
     R_KONST,                       { Realteil der Konstante }
     REAL_ZAHL,                     { Neu errechneter Realteil }
     IMAG_ZAHL,                     { Neu errechneter Imaginärteil }
     REAL_Q,                        { Quadrierter Realteil }
     IMAG_Q       : REAL;           { Quadrierter Imaginäreil }
     STD,MIN,SEC  : INTEGER;        { Variabeln für Rechenzeit }


   {-------------------------------------------------------------------------
    GESTALTUNG DES HINTERGRUNDS UND DER EINGABEMASKE
    -------------------------------------------------------------------------}

   PROCEDURE HINTERGRUND;

    VAR A : INTEGER;

    BEGIN
     CUR_OFF;
     CLEAR_SCREEN;
     HIDE_MOUSE;
     PAINT_STYLE(0);
     PAINT_RECT(1,1,638,398);
     PAINT_STYLE(1);
     PAINT_RECT(4,4,632,392);
     PAINT_STYLE(5);
     PAINT_RECT(5,5,630,390);

     { Gestaltung des Eingabefensters }

     PAINT_STYLE(0);
     PAINT_RECT(113,50,413,65);
     PAINT_STYLE(1);
     PAINT_RECT(116,53,407,59);
     PAINT_STYLE(5);
     PAINT_RECT(117,54,405,57);
     PAINT_STYLE(1);
     PAINT_RECT(125,60,389,45);
     PAINT_STYLE(0);
     PAINT_RECT(123,58,389,45);
     TEXT_STYLE(0);
     TEXT_HEIGTH(13);
```

```
179       DRAW_STRING(235,85,'  Fractale berechnen  ');
180
181       { Gestaltung der Eingabebox }
182
183       PAINT_STYLE(0);
184       PAINT_RECT(113,120,413,246);
185       PAINT_STYLE(1);
186       PAINT_RECT(116,123,407,240);
187       PAINT_STYLE(5);
188       PAINT_RECT(117,124,405,238);
189       PAINT_STYLE(1);
190       PAINT_RECT(165,160,309,156);
191       PAINT_STYLE(0);
192       PAINT_RECT(163,158,309,156);
193       FOR A:=0 TO 3 DO
194        BEGIN
195         PAINT_STYLE(1);
196         PAINT_RECT(172,174+(A*32),143,25);
197         PAINT_STYLE(0);
198         PAINT_RECT(170,172+(A*32),143,25);
199         PAINT_STYLE(1);
200         PAINT_RECT(323,174+(A*32),140,25);
201         PAINT_STYLE(0);
202         PAINT_RECT(321,172+(A*32),140,25);
203        END;
204       GOTOXY(22,11);WRITELN('Realteil Minimum');
205       GOTOXY(22,13);WRITELN('Realteil Maximum');
206       GOTOXY(22,15);WRITELN('Imaginär Minimum');
207       GOTOXY(22,17);WRITELN('Imaginär Maximum');
208      END;
209
210
211  BEGIN
212
213   { Eingabe der Parameter und Fehlerüberprüfung }
214
215   REPEAT
216    HINTERGRUND;
217    CUR_ON;
218    CURSOR(3,0);
219    GOTOXY(42,11);READLN(REAL_MIN);
220    GOTOXY(42,13);READLN(REAL_MAX);
221    GOTOXY(42,15);READLN(IMAG_MIN);
222    GOTOXY(42,17);READLN(IMAG_MAX);
223    IF (REAL_MIN >= REAL_MAX) OR (IMAG_MIN >= IMAG_MAX) THEN
224     BEGIN
225      CUR_OFF;
226      PAINT_STYLE(1);
227      PAINT_RECT(165,160,309,156);
228      PAINT_STYLE(0);
229      PAINT_RECT(163,158,309,156);
230      DRAW_STRING(200,85,'E I N G A B E F E H L E R  !!!');
231      DRAW_STRING(210,200,'Achtung ! Das Minimum darf ');
232      DRAW_STRING(210,230,'nie größer als das Maximum ');
233      DRAW_STRING(210,260,'          sein !           ');
234      I:=SCAN;
235      CUR_ON;
236     END;
237   UNTIL (REAL_MAX > REAL_MIN) AND (IMAG_MAX > IMAG_MIN);
238
239   { Gestaltung des Hintergrunds }
240
241   CUR_OFF;
242   CLEAR_SCREEN;
243   HIDE_MOUSE;
244   PAINT_STYLE(0);
245   PAINT_RECT(1,1,638,398);
```

```
246  PAINT_STYLE(1);
247  PAINT_RECT(4,4,632,392);
248  PAINT_STYLE(5);
249  PAINT_RECT(5,5,630,390);
250
251  { Gestaltung der Eingabebox }
252
253  PAINT_STYLE(0);
254  PAINT_RECT(113,50,413,65);
255  PAINT_STYLE(1);
256  PAINT_RECT(116,53,407,59);
257  PAINT_STYLE(5);
258  PAINT_RECT(117,54,405,57);
259  PAINT_STYLE(1);
260  PAINT_RECT(125,60,389,45);
261  PAINT_STYLE(0);
262  PAINT_RECT(123,58,389,45);
263  TEXT_STYLE(0);
264  TEXT_HEIGTH(13);
265  DRAW_STRING(235,85,'  Fractale berechnen  ');
266  PAINT_STYLE(0);
267  PAINT_RECT(113,120,413,246);
268  PAINT_STYLE(1);
269  PAINT_RECT(116,123,407,240);
270  PAINT_STYLE(5);
271  PAINT_RECT(117,124,405,238);
272  PAINT_STYLE(1);
273  PAINT_RECT(165,160,309,156);
274  PAINT_STYLE(0);
275  PAINT_RECT(163,158,309,156);
276  CUR_ON;
277  GOTOXY(23,11);WRITELN('Bitte wählen Sie einen geeigneten');
278  GOTOXY(23,12);WRITELN('     Wert für die Anzahl der     ');
279  GOTOXY(23,13);WRITELN('          Iterationen !          ');
280  CUR_OFF;
281  TEXT_HEIGTH(4);
282  DRAW_STRING(175,230,'ACHTUNG !!!');
283  DRAW_STRING(175,238,'Wählen Sie für die Anzahl der Iterationen einen');
284  DRAW_STRING(175,246,'kleinen Wert, so ensteht ein grobes Bild ohne  ');
285  DRAW_STRING(175,254,'Feinstruktur. Bei größeren Werten sind Fein-    ');
286  DRAW_STRING(175,262,'strukturen zu erkennen, die aber lange Rechen- ');
287  DRAW_STRING(175,270,'zeiten beanspruchen.(=> Werte zwischen 100 und ');
288  DRAW_STRING(175,278,'350 günstig )                                  ');
289
290  CUR_ON;
291  GOTOXY(37,18);WRITELN('>');
292  REPEAT
293   GOTOXY(38,18);READLN(RECHENTIEFE);
294  UNTIL RECHENTIEFE>0;
295  CUR_OFF;
296  HIDE_MOUSE;
297  CLEAR_SCREEN;
298
299  { Rechenzeit auf Null setzen }
300
301  STD:=0;MIN:=0;SEC:=0;
302  SETTIME(STD,MIN,SEC);
303
304  {------------------------------------------------------------------------
305   DER APFELMÄNNCHEN-ALGORITHMUS
306    ------------------------------------------------------------------------}
307
308  X_KOORDINATE:=400;
309  Y_KOORDINATE:=400;
310
311  { Berechnung der x- und y-Schrittweite }
312
```

```
313   X_SCHRITT:=ABS(((REAL_MAX)-(REAL_MIN))/X_KOORDINATE);
314   Y_SCHRITT:=ABS(((IMAG_MAX)-(IMAG_MIN))/Y_KOORDINATE);
315
316   I_KONST:=IMAG_Min+Y_SCHRITT;
317   FOR Y_Z:=0 TO Y_KOORDINATE DO
318    BEGIN
319     R_KONST:=REAL_MIN+X_SCHRITT;
320     FOR X_Z:=0 TO X_KOORDINATE DO
321      BEGIN
322       REAL_ZAHL:=R_KONST;
323       IMAG_ZAHL:=I_KONST;
324       ZAEHLER:=RECHENTIEFE;
325       REPEAT
326        REAL_Q:=REAL_ZAHL*REAL_ZAHL;
327        IMAG_Q:=IMAG_ZAHL*IMAG_ZAHL;
328        IMAG_ZAHL:=2*REAL_ZAHL*IMAG_ZAHL+I_KONST;
329        REAL_ZAHL:=REAL_Q-IMAG_Q+R_KONST;
330        ZAEHLER:=ZAEHLER-1;
331       UNTIL (REAL_Q+IMAG_Q>4) OR (ZAEHLER=0);
332       IF ODD(ZAEHLER) OR (ZAEHLER=0)THEN PUT_PIXEL(X_Z,Y_Z,1);
333       R_KONST:=R_KONST+X_SCHRITT;
334      END;
335    I_KONST:=I_KONST+Y_SCHRITT;
336   END;
337
338   { Parameterbeschriftung }
339
340   GOTOXY(55,3 );WRITELN('Realteil:');
341   GOTOXY(55,4 );WRITELN('Minimum: ',REAL_MIN:8:8);
342   GOTOXY(55,5 );WRITELN('Maximum: ',REAL_MAX:8:8);
343   GOTOXY(55,7 );WRITELN('Imaginärteil:');
344   GOTOXY(55,8 );WRITELN('Minimum: ',IMAG_MIN:8:8);
345   GOTOXY(55,9 );WRITELN('Maximum: ',IMAG_MAX:8:8);
346   GOTOXY(55,11);WRITELN('Rechentiefe: ',RECHENTIEFE);
347   GETTIME(STD,MIN,SEC);
348   GOTOXY(55,13);WRITELN('Rechenzeit:  ',STD,':',MIN,':',SEC);
349   GET_SCREEN(BILDSCHIRM); { Aktueller Bildschirminhalt wird im
350                             Bildschirmspeicher abgelegt }
351   BILD_SPEICHER:=TRUE;
352   I:=SCAN;
353   SHOW_MOUSE;
354  END;
355
356  {--------------------------------------------------------------------------
357   AUSGABE DER AKTUELLEN PARAMETER
358   --------------------------------------------------------------------------}
359
360  PROCEDURE PARAMETER;
361
362   BEGIN
363    IF (BILD_SPEICHER = FALSE) THEN
364     BEGIN
365      CUR_OFF;
366      CLEAR_SCREEN;
367      HIDE_MOUSE;
368      PAINT_STYLE(0);
369      PAINT_RECT(1,1,638,398);
370      PAINT_STYLE(1);
371      PAINT_RECT(4,4,632,392);
372      PAINT_STYLE(5);
373      PAINT_RECT(5,5,630,390);
374      PAINT_STYLE(1);
375      PAINT_RECT(232,189,194,34);
376      PAINT_STYLE(0);
377      PAINT_RECT(228,185,190,30);
378      GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
379     END
```

```
  ELSE
   PUT_SCREEN(BILDSCHIRM);
   I:=SCAN;
 END;

{-------------------------------------------------------------------------
 FRACTALE SPEICHERN UND LADEN
 -------------------------------------------------------------------------}

PROCEDURE IM_EXPORT;

 VAR FEHLER     : ERR_CODE;      { Fehler-Code beim Laden/Speichern }
     PFAD       : STRING;        { Pfad-Name der Datei              }
     NAME       : STRING;        { Name der Datei                   }
     D          : INTEGER;

BEGIN

 REPEAT

  { Gestaltung des Hintergrunds }

  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);

  { Gestaltung der Eingabebox }

  PAINT_STYLE(0);
  PAINT_RECT(213,100,213,65);
  PAINT_STYLE(1);
  PAINT_RECT(216,103,207,59);
  PAINT_STYLE(5);
  PAINT_RECT(217,104,205,57);
  PAINT_STYLE(1);
  PAINT_RECT(225,110,189,45);
  PAINT_STYLE(0);
  PAINT_RECT(223,108,189,45);
  TEXT_STYLE(0);
  TEXT_HEIGTH(13);
  DRAW_STRING(235,125,'  Erstellte Bilder  ');
  DRAW_STRING(235,144,'Im- bzw. Exportieren');

  { Unterer Kasten in der Mitte }

  PAINT_STYLE(0);
  PAINT_RECT(213,170,213,150);
  PAINT_STYLE(1);
  PAINT_RECT(216,173,207,144);
  PAINT_STYLE(5);
  PAINT_RECT(217,174,205,142);
  FOR D:=0 TO 2 DO BEGIN
   PAINT_STYLE(0);
   PAINT_RECT(220,185+(D*42),40,39);
   PAINT_STYLE(1);
   PAINT_RECT(225,190+(D*42),32,31);
   PAINT_STYLE(0);
   PAINT_RECT(223,188+(D*42),32,31);
   PAINT_RECT(265,185+(D*42),154,39);
   PAINT_STYLE(1);
   PAINT_RECT(270,190+(D*42),146,31);
   PAINT_STYLE(0);
   PAINT_RECT(268,188+(D*42),146,31);
```

```
  END;

  { Beschriftung der Menüboxen }

  TEXT_STYLE(0);
  TEXT_HEIGTH(13);
  DRAW_STRING(227,208,'F 6');
  DRAW_STRING(227,250,'F 7');
  DRAW_STRING(227,292,'ESC');
  TEXT_HEIGTH(4);
  DRAW_STRING(278,200,'   Erstellte Bilder   ');
  DRAW_STRING(278,209,'importieren bzw. laden');
  DRAW_STRING(278,238,'   Erstellte Bilder   ');
  DRAW_STRING(278,246,'   exportieren bzw.   ');
  DRAW_STRING(278,254,'      speichern       ');
  DRAW_STRING(278,288,' Zurück ins Hauptmenü ');
  I:=SCAN;
  IF I=64 THEN
   BEGIN
    SHOW_MOUSE;
    NAME:='';
    PFAD:='A:\*.*';
    IF GET_IN_FILE(PFAD,NAME) THEN
     BEGIN
      CLEAR_SCREEN;
      FEHLER:=READ_SCREEN(1,NAME,BILDSCHIRM);
      PUT_SCREEN(BILDSCHIRM);
      BILD_SPEICHER:=TRUE;
      I:=SCAN;
     END;
   END;
  IF I=65 THEN
   BEGIN
    IF BILD_SPEICHER = FALSE THEN
     BEGIN
      CUR_OFF;
      CLEAR_SCREEN;
      HIDE_MOUSE;
      PAINT_STYLE(0);
      PAINT_RECT(1,1,638,398);
      PAINT_STYLE(1);
      PAINT_RECT(4,4,632,392);
      PAINT_STYLE(5);
      PAINT_RECT(5,5,630,390);
      PAINT_STYLE(1);
      PAINT_RECT(232,189,194,34);
      PAINT_STYLE(0);
      PAINT_RECT(228,185,190,30);
      GOTOXY(30,12);WRITELN('KEIN BILD IM SPEICHER');
      I:=SCAN;
     END
    ELSE
     BEGIN
      SHOW_MOUSE;
      CLEAR_SCREEN;
      PARAMETER;
      NAME:='';
      PFAD:='A:\*.*';
      IF GET_IN_FILE(PFAD,NAME) THEN
       BEGIN
        FEHLER:=WRITE_SCREEN(1,NAME,BILDSCHIRM);
       END;
     END;
   END;
 UNTIL I=1;
END;
```

```
{------------------------------------------------------------------------
 INFORMATION
 ------------------------------------------------------------------------}

PROCEDURE INFO;

 BEGIN
  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);
  PAINT_STYLE(0);
  PAINT_RECT(50,40,540,320);
  PAINT_STYLE(1);
  PAINT_RECT(60,50,520,300);
  PAINT_STYLE(0);
  PAINT_RECT(58,48,520,300);
  TEXT_HEIGTH(13);
  TEXT_STYLE(0);
  DRAW_STRING(70,70,'Information');
  DRAW_STRING(70,110,'Das vorliegende Computerprogramm ermöglicht die graphische');
  DRAW_STRING(70,130,'Darstellung der Mandelbrot-Menge. Hierzu müssen in F2 fünf');
  DRAW_STRING(70,150,'Eingabegrößen festgelegt werden. Bitte beachten Sie bei der');
  DRAW_STRING(70,170,'Eingabe,daß Sie keine Buchstaben eingeben. Verwenden Sie ');
  DRAW_STRING(70,190,'für Kommazahlen einen Punkt. Bei einer falschen Eingabe kann');
  DRAW_STRING(70,210,'durchaus vorkommen,daß das Programm abstürzt. Dieses schwer-');
  DRAW_STRING(70,230,'wiegende Problem konnte nicht gelöst werden, da es sich um');
  DRAW_STRING(70,250,'eine aufwendige Routine handelt ( Die Eingabe müßte als String');
  DRAW_STRING(70,270,'deklariert werden und auf jede Stelle untersucht werden ).');
  DRAW_STRING(70,290,'Dieses Programm wurde in Pascal erstellt. Pascal ist im Gegen-');
  DRAW_STRING(70,310,'satz zu Assembler relativ langsam. Dennoch wünsche ich dem ');
  DRAW_STRING(70,330,'Benutzer beim Erstellen von Fractale viel Spaß. (TASTE !)');
  I:=SCAN;
 END;

{------------------------------------------------------------------------
 HAUPTMENÜ
 ------------------------------------------------------------------------}
PROCEDURE HAUPTMENU;

 VAR A : INTEGER;

 BEGIN

  { Gestaltung des Hintegrunds }

  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);

  { Fenster in der Mitte }

  PAINT_STYLE(0);
  PAINT_RECT(182,20,280,360);
  PAINT_RECT(184,22,276,356);
  PAINT_STYLE(4);
  PAINT_RECT(185,23,274,354);

  { Hintergrund der Menübox }

```

```
  PAINT_STYLE(0);
  PAINT_RECT(190,28,264,344);
  PAINT_RECT(192,30,260,340);
  PAINT_RECT(193,31,258,338);
  PAINT_STYLE(5);
  PAINT_RECT(194,32,256,336);

  { Beschriftung der Menübox }

  PAINT_STYLE(0);
  PAINT_RECT(199,40,245,55);
  PAINT_RECT(201,42,241,51);
  TEXT_STYLE(1);
  TEXT_HEIGTH(13);
  DRAW_STRING(250,65,'MANDELBROT - MENGEN');
  TEXT_HEIGTH(4);
  DRAW_STRING(220,78,'Das Computerprogramm zur Facharbeit');
  DRAW_STRING(220,88,'         von Urban Bettag          ');
  FOR A:=1 TO 6 DO
   BEGIN
    PAINT_STYLE(0);
    PAINT_RECT(199,79+(A*32),50,29);PAINT_RECT(279,79+(A*32),146,29);
    PAINT_STYLE(1);
    PAINT_RECT(204,84+(A*32),42,21);PAINT_RECT(284,84+(A*32),138,21);
    PAINT_STYLE(0);
    PAINT_RECT(202,82+(A*32),42,21);PAINT_RECT(282,82+(A*32),138,21);
   END;
  TEXT_STYLE(0);
  TEXT_HEIGTH(13);
  DRAW_STRING(212,130,'F 1');TEXT_HEIGTH(4);DRAW_STRING(288,122,'   Erstelle Bilder  ');
                                            DRAW_STRING(292,130,'Im- bzw. Exportieren');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,162,'F 2');TEXT_HEIGTH(4);DRAW_STRING(292,158,' Fractale berechnen ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,194,'F 3');TEXT_HEIGTH(4);DRAW_STRING(292,186,' Aktuelle Parameter ');
                                            DRAW_STRING(292,193,'     einblenden     ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,226,'F 4');TEXT_HEIGTH(4);DRAW_STRING(292,222,' Fractal ausdrucken ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,258,'F 5');TEXT_HEIGTH(4);DRAW_STRING(274,294,'  Information für   ');
                                            DRAW_STRING(292,258,'     Benutzer       ');
  TEXT_HEIGTH(13);
  DRAW_STRING(212,290,'F10');TEXT_HEIGTH(4);DRAW_STRING(292,286,'  Programm beenden  ');
  TEXT_HEIGTH(13);
 END;

{------------------------------------------------------------------------
 BEENDEN
 ------------------------------------------------------------------------}

 PROCEDURE BEENDEN;

 BEGIN
  CLEAR_SCREEN;
  PAINT_STYLE(0);
  PAINT_RECT(1,1,638,398);
  PAINT_STYLE(1);
  PAINT_RECT(4,4,632,392);
  PAINT_STYLE(5);
  PAINT_RECT(5,5,630,390);
  PAINT_STYLE(1);
  PAINT_RECT(210,180,220,40);
  PAINT_STYLE(0);
  PAINT_RECT(208,178,220,40);
  DRAW_STRING(220,200,'Sind Sie sicher (J/N) ?');
  I:=SCAN;
  IF I=36 THEN EN:=FALSE;
```

```
  IF I=49 THEN EN:=TRUE;
 END;

{------------------------------------------------------------------------
 HAUPTPROGRAMM
 ------------------------------------------------------------------------}

BEGIN

 { GEM - Routinen werden aufgerufen und instaliert

   Init_Gem
   --------

   Die Funktion führt alle notwendigen Operationen durch, um GEM mitzuteilen
   daß nun eine Applikation an den Segnungen des Betriebssystems teilhaben
   möchte. Tritt ein Fehler auf so liefert die Funktion Init_gem der Wert -1
   Eine erfolgreiche Operation liefert immer einen positiven Wert.

   Exit_Gem
   --------

   Beendet ein GEM - Programm (Graphic Environment Manager). Die Prozedur
   teil GEM mit, daß ab nun keine weiteren GEM-Aufrufe getätigt werden.
   Bei einem fehlerhaften Aufruf von Init_Gem sollte man diese Prozedur nicht
   aufrufen. }

 IF INIT_GEM >=0 THEN
  BEGIN
   BILD_SPEICHER:=FALSE;
   EN:=TRUE;
   REPEAT
    HAUPTMENU;
    I:=SCAN;
    IF I=59 THEN IM_EXPORT;
    IF I=60 THEN BERECHNEN;
    IF I=61 THEN PARAMETER;
    IF I=62 THEN DRUCKEN;
    IF I=63 THEN INFO;
    IF I=68 THEN BEENDEN;
    SHOW_MOUSE;
   UNTIL EN=FALSE;
   EXIT_GEM;
  END;
END.
```

Notes on the program

- **Historical findings:** The program was evidently written for an Atari ST computer with the GEM interface and BIOS/XBIOS calls. Identifiers such as `TEXT_HEIGTH` have not been modernised, because they may come from the missing libraries.
- **Scan code:** The return value of `BCONCIN` contains the scan code in the upper word. This would call for a division by $65536 = 2^{16}$ or a shift by 16 bits. The printout reads `DIV (65535)`; this is very probably an error in the original program and has been deliberately retained.
- **Menu label:** In the main menu, the label “Information für” is at $y = 294$, and the corresponding second line “Benutzer” at $y = 258$. The two words therefore appear separately on the screen; this too is an error in the original and is preserved.

- **Grid bounds:** The loops each run from 0 up to and including 400, that is, over 401 values. At the same time, the real and imaginary parts are increased by one step before the first pixel. As a result, the stated minima are skipped and values above the maxima are reached at the end. In addition, the y-coordinate 400 may lie outside the intended grid for a screen area from 0 to 399.
- **Iteration:** For each parameter c, the computation in the program starts with $z = c$, not with $z_0 = 0$. For the pure escape test this amounts to a count shifted by one step; for the black-and-white colouring used, however, it changes the parity of the iteration count. Moreover, the termination sum `REAL_Q+IMAG_Q` is formed from the squares before the update just computed.
- **Representation:** Exterior points are drawn black or white according to the parity of the remaining counter. The stripes therefore show escape-time classes. They are not in themselves proof that a point belongs to the Mandelbrot set.
- **Present-day assessment:** With the missing GEM/TRIX files and a suitable historical Pascal system, the syntax could be checked more precisely. Without these components, any supposedly runnable reconstruction would be speculative; for this reason only the program text actually available has been included here.

10.5 Python post-processing from 2026

The following Python programs are not part of the original Facharbeit of 1990. They were written in 2026 for the digital revision. The first program computes the new black-and-white, colour and detail images of the Mandelbrot set. The second program produces the three-dimensional escape-time representation. Both programs use the mathematical iteration $z_{n+1} = z_n^2 + c$ with the starting value $z_0 = 0$.

In each case only the computational core is printed; the line numbers correspond to the complete file. The remaining parts of the programs merely produce labels and the image layout. The complete scripts `generate_fractals.py`, `generate_3d_fractal.py`, `generate_diagrams.py` and `generate_ifs.py` are not printed.

Escape times for the two-dimensional images

`generate_fractals.py`: function `escape_times`

```
40 def escape_times(bounds, iterations, width=1000):
41     """Berechne die erste Fluchtiteration fuer z_0=0, z_(n+1)=z_n^2+c."""
42     xmin, xmax, ymin, ymax = bounds
43     height = max(2, round(width * (ymax - ymin) / (xmax - xmin)))
44     x = np.linspace(xmin, xmax, width, dtype=np.float64)
45     y = np.linspace(ymin, ymax, height, dtype=np.float64)
46     c = x[np.newaxis, :] + 1j * y[:, np.newaxis]
47     z = np.zeros_like(c)
48     active = np.ones(c.shape, dtype=bool)
49     escaped = np.full(c.shape, iterations, dtype=np.uint16)
50 
51     for n in range(1, iterations + 1):
52         z[active] = z[active] * z[active] + c[active]
53         newly = active & (z.real * z.real + z.imag * z.imag > 4.0)
54         escaped[newly] = n
55         active[newly] = False
56         if not active.any():
57             break
58     return escaped
```

For each pixel c, the sequence is iterated from $z_0 = 0$ until $|z_n|^2 > 4$ holds or the iteration limit is reached. What is stored is the index of the first crossing, that is, the escape time.

Height field for the three-dimensional representation

`generate_3d_fractal.py`: function `height_field`

```
44 def height_field(nx=230, ny=190):
45     xmin, xmax, ymin, ymax = BOUNDS
46     xs = np.linspace(xmin, xmax, nx)
47     ys = np.linspace(ymin, ymax, ny)
48     c = xs[np.newaxis, :] + 1j * ys[:, np.newaxis]
49     z = np.zeros_like(c)
50     active = np.ones(c.shape, dtype=bool)
51     smooth = np.full(c.shape, float(MAX_ITER), dtype=np.float64)
52 
53     for iteration in range(1, MAX_ITER + 1):
54         z[active] = z[active] * z[active] + c[active]
55         radius2 = z.real * z.real + z.imag * z.imag
56         escaped = active & (radius2 > 4.0)
```

```
        if escaped.any():
            modulus = np.sqrt(radius2[escaped])
            smooth[escaped] = (
                iteration + 1.0 - np.log(np.log(modulus)) / np.log(2.0)
            )
        active[escaped] = False
        if not active.any():
            break

    # Schnelles Entweichen ergibt grosse Hoehen. Nicht entwichene Punkte sind
    # der ebene See auf Hoehe null. Die Wurzel macht die Kueste besser sichtbar.
    height = np.zeros_like(smooth)
    outside = ~active
    speed = np.clip((MAX_ITER - smooth[outside]) / MAX_ITER, 0.0, 1.0)
    height[outside] = 0.20 + 1.85 * np.sqrt(speed)
    return xs, ys, height, active
```

Here the escape time is smoothed and converted into a height: the faster a point escapes, the higher it lies. Points that have not escaped form the flat surface of the “Mandelbrot lake”.

Change log

The table lists the most important corrections of content on original pages 20 to 24. It is not exhaustive.

Page	Original problem	Correction	Effect
20	Starting value and index shifted	$z_0 = 0$, $z_1 = k$, $z_2 = k^2 + k$	Recursion and program count agree
21	Parameter k drops out	Case distinction explicitly restricted to $k = 0$	The existing section is kept as a special case
22	$\mathcal{M}$ defined via convergence to zero	Boundedness of the critical orbit	Correct definition of the set
22–23	Finite test taken as proof of membership	Non-escape up to N as an approximation	The algorithmic statement becomes precise
23	New components not written back	Update a and b after every step	The pseudocode actually performs an iteration
24	Parameter plane and starting-value plane conflated	Separate the Mandelbrot and Julia planes	The relationship between the two sets is described correctly

Page numbers refer to the 1990 original.

Source basis

The basis is the original scans `IMG_1917` to `IMG_1960`, covering printed pages 1 to 44. File number `IMG_1950` is missing; according to the author, the corresponding gap is closed by the colour plate `IMG_1960`, photographed later. For the partial check of the program, the visible program core in `IMG_1965` was also used. The historical corrections use mathematical foundations that were already available in 1990; later findings are explicitly marked as such.